Technische Universität München · Hochschule für Politik München

Ethnic Signals and Investor Choice

Municipal Ethnic Composition and Inward FDI in Bosnia and Croatia during EU Accession

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1 Introduction

In 2001, Croatia signed a Stabilisation and Association Agreement (SAA) with the European Union (Stabilisation and Association Agreement with Croatia 2001). This agreement is the framework that sets a prospective member on the path toward accession. Bosnia and Herzegovina (hereafter Bosnia) followed suit in 2008 (Stabilisation and Association Agreement with Bosnia and Herzegovina 2008). Two neighbors and former Yugoslav republics, moving through the same accession framework toward the same destination within the same decade, did not arrive together. Croatia joined the EU in 2013 (Treaty of Accession of Croatia 2011). Nearly two decades after signing its own agreement, Bosnia has not. It reached candidate status only in 2022 (European Council 2022). Enlargement is usually studied in terms of pace and degree, meaning how far a country integrates, how quickly, and how much foreign investment its progress attracts. Far less examined is whether accession shapes not the quantity of that investment but its kind. If moving toward Europe reshapes the character of the foreign capital a country attracts, favoring production geared toward export over investment that serves the local market, then two states that set out from the same point and ended at different ones should, over time, host visibly different profiles of foreign investment. Such a difference should appear not only between the two states but also within them, between the places most and least firmly oriented toward Europe.

The distinction between quantity and type matters because foreign direct investment (FDI) is not a single thing. Multinational firms invest for different reasons, and these reasons shape which FDI a host economy receives. Market-seeking investment locates production to serve the local market. Its scale typically follows domestic demand, and it need not bring the investor's most advanced technology. Export-oriented investment, by contrast, places production wherever costs are lowest to sell into global markets. It requires the firm to transfer its most productive, firm-specific assets (Pandya 2016, 469). The two types also carry different payoffs. Export-oriented investments create high-quality jobs, introduce advanced technologies, and generate technological spillovers, all without increasing competition in the domestic product market (Pandya 2016, 456–57, 469). Countries therefore have stronger reasons to court this type of foreign capital (Pandya 2016, 457). Market-seeking investments tend to bring fewer of these gains because they serve domestic demand, compete directly with local firms, and can rely on older production technologies (Pandya 2016, 457, 469). What a country gains from foreign investment depends on the composition of what it draws in, not simply on its volume. Because export-oriented production is freer to locate wherever conditions favor it, it is also more sensitive to the environment it enters (Pandya 2016, 457, 469).

If the type of investment a country attracts depends on its environment, the next question is which features of that environment matter. So far, the dominant answer has been national. Political conditions can sort foreign investment by type, but the conditions studied belong to the nation as a whole, such as the treaties a state signs or the party that governs it (Kerner and Lawrence 2014; Pinto and Pinto 2008). Ethnicity enters this tradition as one item on a national list of risks to the volume of investment (Busse and Hefeker 2007), or as an index that records how divided a society is but not what its groups stand for (Montalvo and Reynal-Querol 2005a, 2005b). Research on EU accession shares the national frame. Accession announcements drew investment into candidate countries (Bevan and Estrin 2004; Clausing and

Dorobantu 2005), but each milestone is treated as reaching the whole country at once. More recently, the literature has moved below the national level. Most of this work asks what foreign investment does once it arrives, from raising local economic activity to lowering pre-election violence in competitive regions (Rommel, Palmtag, and Messerschmidt 2025; Palmtag, Paula, and Rommel 2025). A smaller body of work makes location the outcome and, in Mexico, shows that local politics shapes the type of investment as well (Samford and Ortega Gómez 2014). Where ethnicity appears in this work, it is a network that ties a place to one foreign country (Burchardi, Chaney, and Hassan 2019), not a signal about the place itself. None of these literatures asks whether the ethnic composition of a municipality, read as a signal of its orientation toward the EU, predicts the type of FDI it attracts, or whether that relationship changes as a country moves through accession.

This paper addresses that gap. I ask the following question. Does the ethnic composition of municipalities in Croatia and Bosnia systematically predict the composition of inward FDI, and does this relationship vary as these countries progress through the EU accession process?

The argument begins with what ethnic composition signals. In Bosnia, a municipality's ethnic majority is closely tied to how firmly it is oriented toward the EU. Ethnicity is written into the constitution (General Framework Agreement 1995, annex 4), and the Serb-majority side has an institutional home in Republika Srpska, where significant segments of public opinion reject pro-EU policies (Jović 2018). Ethnic composition is also easy to observe. An investor does not need to study local politics to know which group holds the majority in a municipality. This signal should matter most to export-oriented investors. They transfer their most productive assets and have more freedom in where to locate, so they have both more to lose and more alternatives (Pandya 2016, 457, 469). The first hypothesis follows. In Bosnia, Bosniak-majority municipalities should attract a higher share of export-oriented relative to market-seeking FDI than comparable Serb-majority municipalities.

Accession may then change the strength of this relationship, and the argument runs in two directions. As a country takes on binding commitments, a municipality's orientation toward the EU could matter more, because the question becomes whether reforms are carried out locally or resisted. Yet the same commitments bind the whole state and are monitored from outside, so they could also stand in for the local signal and narrow the differences between municipalities. The second hypothesis therefore expects the relationship to change as accession advances, without predicting the direction.

I test this argument on Bosnia and use Croatia as a benchmark. Croatia completed accession within the period, but none of its Serb-majority municipalities, 18 under the 2011 census and 16 under the 2021 census, records FDI in any year. It therefore shows how FDI developed in Croat-majority municipalities through a completed accession, but it cannot supply an ethnic comparison. The data consists of greenfield project announcements from fDi Markets between 2003 and 2023 (The Financial Times Ltd. 2026). Each project is classified as export-oriented or market-seeking by its sector and business activity and matched to its municipality, which yields 356 projects across 142 Bosnian municipalities and 754 across 556 Croatian ones. Because FDI arrives in only 6.7% of Bosnian municipality-years, the analysis uses a two-part model. The first part asks whether any FDI arrives; the second asks what share of it is export-oriented where it does. Accession enters as a difference-in-differences around Bosnia's SAA in 2008 and its candidacy in 2022.

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The evidence does not support the first hypothesis. Where FDI arrives, Bosniak-majority municipalities do not attract a higher export-oriented share than Serb-majority ones. They do attract FDI more often, but that difference reflects city size, not ethnic composition. Nor does accession change the relationship for FDI type, so the second hypothesis is not supported either. The one change appears in arrival. Before 2008, Bosniak-majority municipalities held a modest advantage in attracting any FDI even at equal size, and after 2008 it disappears. This convergence holds when any single municipality or the five largest FDI destinations are dropped, but the data cannot separate it from a shift between the two entities. In Croatia, completed accession leaves both arrival and type in Croat-majority municipalities unchanged. Because the type model rests on only 201 municipality-years with FDI, its results rule out large effects, not modest ones. In short, the ethnic signal the argument expects does not show in the type of FDI, and the one difference in whether FDI arrives at all is gone after 2008.

The paper proceeds as follows. Section 2 reviews the literature on why the type of FDI matters, on political risk and ethnicity, on EU accession, and on the subnational turn, and it sets out how ethnicity and the EU are linked in Bosnia and Croatia. Section 3 develops the argument that ethnic composition signals a municipality's orientation toward the EU, derives the two hypotheses, and sets the scope of the argument for Croatia. Section 4 describes how the FDI data is classified and matched to municipalities, the two-part model, the difference-in-differences design, and the main threats to identification. Section 5 presents the descriptive patterns, the results for arrival and type, the accession tests and their robustness, and the Croatian benchmark. Section 6 discusses what the results mean, including the role of city size, the difference between whether FDI arrives and what kind arrives, and the overlap between ethnicity and entity. Section 7 concludes.

2 Literature Review

This section reviews the literature on which the paper builds and shows where it stops. It proceeds in five steps. First, it explains why the type of FDI matters. Second, it turns to political risk and the place of ethnicity within it. Third, it reviews how EU accession shapes foreign investment. Fourth, it covers the recent move toward subnational analysis of FDI. Fifth, it sets out how ethnicity and the EU are linked in Bosnia and Croatia. The section closes by drawing these strands together and identifying the gap this paper addresses.

2.1 Why the Type of FDI Matters

The political economy of FDI has long treated foreign investment as a single quantity. Its central questions concern how much investment a country attracts and which political conditions raise or lower that amount. In her review of the field, Pandya (2016, 466) notes that, until recently, scholars had "not explored in depth the political economy implications" of the variation within FDI. She argues that research can advance "by disaggregating FDI into multinational firms' specific production activities," because the scope for countries to compete for investment and firms' sensitivity to property rights "vary widely" (Pandya 2016, 455). Kerner (2014) makes a related point about measurement. He shows that the FDI data most often used in political science combines activities that are conceptually distinct, and the relationship between democracy, political risk, and multinational behavior changes once the measure matches the theory. In short, FDI is not one phenomenon, and treating it as one obscures how politics shapes it.

The most useful division separates FDI that substitutes for trade from FDI that complements it (Pandya 2016, 466). The first is market-seeking, where a firm produces in a country to sell in that country. The second is export-oriented, where a firm produces in one country to sell in others, often as one stage of a production network that spans several countries. The division matters because "depending on the type of FDI in question, countries have qualitatively different incentives for FDI openness, as do MNCs in their location decisions" (Pandya 2016, 466). It also changes what drives location. Before export-oriented production spread, "immutable location-specific characteristics such as market size and resource endowments drove most FDI location choices" (Pandya 2016, 466). For market-seeking investment, such fixed characteristics still weigh heavily (Pandya 2016, 457). Vertically integrated production networks, however, gave multinationals more flexibility in where to place export-oriented projects and allowed countries to compete for them (Pandya 2016, 466).

A smaller body of work traces how politics shapes the composition of FDI rather than only its volume. Kerner and Lawrence (2014) argue that the political risk attached to FDI is not a uniform property of FDI but a function of investment in fixed capital. They find that bilateral investment treaties with the United States correlate with fixed-capital investment and show little relationship with other measures of multinational activity. Pinto and Pinto (2008) find that, across OECD countries, foreign investment flows into different sectors of the host economy under left- and right-leaning governments. Both studies show that political conditions can sort investment by type. Both, however, locate those conditions in two attributes of the nation, the treaties a state signs and the party that governs it. The type of FDI a country receives is explained by what the country is, not by where within it the investment goes. The literature on political risk, to which ethnicity belongs, shares this national frame.

2.2 Political Risk, Ethnicity, and FDI

The largest body of work on politics and FDI asks which national conditions attract investment. Democracies receive more FDI than autocracies, both across countries and within countries over time (Jensen 2003). Li and Resnick (2003) qualify this finding. They show that property rights protection drives the relationship and that, once its effect is held constant, democratic institutions reduce FDI inflows. Jensen (2008) measures political risk more directly, through the premiums multinationals pay for political risk insurance, and finds that democratic regimes lower that risk mainly by constraining the executive. Busse and Hefeker (2007) widen the lens to twelve indicators of political risk and institutional quality across 83 developing countries between 1984 and 2003. They find that government stability, the absence of internal conflict and ethnic tensions, basic democratic rights, and law and order are highly significant determinants of FDI inflows. In this tradition, political risk is an attribute of the host country, and the outcome is how much investment arrives. Ethnicity enters as one item on a national list.

A separate literature in economics treats ethnicity as a source of costs. Easterly and Levine (1997) show that ethnic diversity helps explain cross-country differences in public policy, and that Africa's high ethnic fragmentation explains a significant part of its political instability and poor economic outcomes. Montalvo and Reynal-Querol (2005a, 2005b) argue that an index of polarization, rather than fractionalization, captures the potential for conflict. They find that ethnic polarization predicts the incidence of civil war and reduces economic development, in part through lower investment. Alesina and La Ferrara (2005) survey this work and name the measurement of diversity and the endogenous formation of ethnic identity as open issues. These

studies share a design choice. They summarize a country's ethnic structure in a single index. Such an index records how divided a society is. It does not record which group holds the majority or what that group stands for politically. In this literature, ethnicity carries no direction.

Two qualifications follow from this work. First, the cost of diversity depends on institutions. Easterly (2001) finds that where institutions are good enough, ethnic diversity neither lowers growth nor worsens economic policy. Second, the effect of ethnicity depends on scale. Montalvo and Reynal-Querol (2021) note that cross-country studies find a negative or insignificant relationship between diversity and growth, while studies of cities find positive effects on wages and productivity. Using cells of fixed size, they find a positive relationship between diversity and growth in small geographic areas. Closer to the cases at hand, Brada, Kutan, and Yigit (2006) find that conflict and instability pushed FDI to the Balkans below the level expected for comparable West European countries. This literature thus establishes that ethnic division, and the instability tied to it, can deter investment. It does so at the national level, with ethnicity measured as undirected division and with the volume of investment or growth as the outcome. It also suggests that the answer may change with the quality of institutions and with the level at which ethnicity is measured.

2.3 EU Accession as a Credibility Anchor

A large body of research asks whether international agreements help states make credible commitments to foreign investors. Bilateral investment treaties and preferential trade agreements "address the obsolescing-bargain problem by enshrining openness in international law" (Pandya 2016, 464). Whether such agreements can make up for weak domestic institutions, however, "remains contested" (Pandya 2016, 464). EU accession is the most far-reaching commitment of this kind available to a European state. A candidate does not sign a single treaty but takes on the obligations of membership step by step.

Research on the eastern enlargement finds that this commitment drew investment. Bevan and Estrin (2004) study bilateral FDI from Western, mainly EU, countries into Central and Eastern Europe. They find that unit labor costs, gravity factors, market size, and proximity are the most important influences, and host country risk is not significant. Announcements about EU accession, however, affected FDI into the future member states. Clausing and Dorobantu (2005) reach a similar conclusion. Alongside the traditional market size and cost factors, they find that key EU announcements on the accession process had statistically significant and quantitatively important effects on FDI into the candidate countries. Bandelj (2010) questions the mechanism. Earlier studies held that the prospect of membership attracted FDI by lowering perceived risk. She shows instead that accession worked indirectly, by shaping how strongly post-socialist states promoted FDI, and that this promotion raised inflows, net of conventional risk and return factors. Accession, then, raised investment, but whether it did so by lowering risk or by changing what states did remains open.

The promise of membership does not close every gap. Estrin and Uvalic (2014) compare FDI across transition economies and find that the Western Balkan countries in their sample, Bosnia among them, receive less FDI than other transition economies, even after accounting for market size, distance from source countries, institutional quality, and prospects of EU membership. They also document differences in the sectoral structure of FDI across the region, again country by country. This literature shares two features. First, its unit is the country, and an accession milestone reaches all of a candidate at once. Second, its outcome is mostly the volume of

investment. None of these studies asks whether accession changes the differences between places within a candidate country, or the mix of investment those places attract. Answering that question requires data below the national level.

2.4 The Subnational Turn

Over the past decade, research on FDI has moved below the national level. Using geocoded project data, it maps foreign investment to specific places and asks what that investment does there. FDI raises local economic activity around project sites and widens regional inequality, more so in autocracies than in democracies (Rommel, Palmtag, and Messerschmidt 2025). It shapes politics as well. A newly announced greenfield project raises the probability that the incumbent party wins reelection in Brazilian mayoral elections (Owen 2019). In autocracies, FDI bolsters regime support among the well educated and lowers it among the poorly educated (Rommel 2024). In competitive regions of sub-Saharan Africa, it reduces the violence incumbents use before elections (Palmtag, Paula, and Rommel 2025). A related body of work traces social costs. FDI in all sectors increases local conflict, although only extractive FDI escalates toward civil war (Brazys, de Soysa, and Vadlamannati 2025). Industrial mining increases bribe payments (Knutsen et al. 2017) and protests, but not rebel activity (Christensen 2019). FDI, by contrast, reduces individual experiences of bribery where existing corruption is high (Brazys and Kotsadam 2020).

For these studies, where FDI goes is a problem of identification rather than a question. Investment does not arrive at random, so the places that receive it may differ from those that do not. The designs respond by narrowing the comparison. Rommel, Palmtag, and Messerschmidt (2025, 6) draw buffers around project sites so that they "only compare locations that are at some point in time attractive for multinational enterprises." Rommel (2024) compares respondents already exposed to FDI with those not yet exposed. Owen (2019, 628) matches municipalities on variables drawn from the site selection literature. Knutsen et al. (2017, 320) find that mines are initially located in less corrupt areas and become more corrupt only after they open. Palmtag, Paula, and Rommel (2025, 253) argue that once investors have chosen a country, logistical needs such as access to resource deposits or trade routes restrict where within it they can locate, and they find no statistically significant effect of pre-election violence on regional FDI inflows. These are sound choices for the questions these studies ask. They do mean, however, that investors' choice of place within a country is treated as a threat to identification, to be controlled or tested, rather than as the outcome to be explained.

A smaller literature makes subnational location the outcome. Samford and Ortega Gómez (2014) model FDI across the 32 Mexican states and divide it into resource-, market-, and efficiency-seeking investment. Local party control, social stability, and the perceived effectiveness of state authorities all predict where investment goes, and each type responds to distinct social, political, and economic dynamics. Garriga (2022) finds that investors in Mexico prefer states run by left-wing governors. Ledyaeva, Karhunen, and Kosonen (2013) show that investors from less corrupt and more democratic countries locate in less corrupt and more democratic Russian regions, while investors from more corrupt and less democratic countries do the opposite. Hunnicutt (2024) argues that deployments of UN police credibly signal to foreign firms where instability is least likely to disrupt their operations, and finds that they encourage new extractive concessions in post-conflict Liberia. Political conditions within a country, then, do shape where investment goes, and in Mexico they shape its type.

Where ethnicity enters this work, it does so as a network. Burchardi, Chaney, and Hassan (2019) show that the ancestry composition of US counties has a causal effect on FDI between local firms and the ancestral countries, and that the effect runs mainly through lower information frictions. Ethnicity here links a place to one specific foreign country. It does not tell investors from anywhere about the political direction of the place itself. The subnational literature has shown that FDI changes the places it reaches and that local politics can steer it. It has not asked whether the ethnic composition of a place, read as a signal of its political orientation, shapes the type of investment that place attracts.

2.5 Ethnicity, Entity, and Europe in Bosnia and Croatia

In Bosnia, ethnicity is not only a social fact but a constitutional one. The Constitution, Annex 4 of the Dayton Agreement, names "Bosniacs, Croats, and Serbs, as constituent peoples (along with Others)," divides the state into two Entities, the Federation of Bosnia and Herzegovina and Republika Srpska, and allows a proposed decision of the Parliamentary Assembly to be challenged as "destructive of a vital interest of the Bosniac, Croat, or Serb people" (General Framework Agreement 1995, annex 4, preamble, arts. I(3), IV(3)(e)). Politics follows this design below the national level. Kapidžić (2020, 81) finds that "the political contest in BiH is purposefully contained within ethnic and subnational boundaries," and that the drift toward illiberal rule is unevenly distributed across subnational units. Hadzic, Carlson, and Tavits (2020) show that communities more exposed to wartime violence developed stronger ethnic identities and greater distrust of other groups, which made ethnic parties the most attractive channels of representation. Ethnicity in Bosnia is therefore institutional and local at once.

The same divisions shape Bosnia's path toward the EU. Vasilev (2011) compares EU conditionality in Bosnia and Macedonia and finds that ethnic preferences hardened in Bosnia in the course of accession politics, while they were transformed in Macedonia. Sebastian (2009) traces the failed constitutional reform of 2005 and 2006 and attributes it both to irreconcilable domestic differences and to the weak leverage of the EU, despite its assumed power of conditionality. Bieber (2018) argues that competitive authoritarian practices returned across the region in part because the transformative power of external actors, first and foremost the EU, weakened. Public opinion follows the same lines, with significant segments in Republika Srpska rejecting pro-EU policies (Jović 2018). Accession has not dissolved ethnic politics in Bosnia. Ethnic and entity identity continue to shape how places relate to the EU path.

Croatia offers a contrast. It completed accession as a single state, and its Serb population is a minority without an entity of its own. Ethnicity still matters locally. Glaurdić, Mochtak, and Lesschaeve (2023) find that ethnic bias in Croatian voting depends on the local balance between groups and on exposure to wartime violence, particularly in communities populated by the Serb minority. In both countries, then, ethnic composition shapes politics at the local level. This literature, however, is written in terms of elections, institutions, and conditionality. It does not ask what these divisions communicate to foreign investors.

2.6 The Gap

Taken together, the strands reviewed here leave a specific question open. First, the type of FDI matters, and politics can shape it, but existing explanations place that politics in attributes of the nation. Second, ethnicity enters the political risk literature as a national, undirected measure of division that predicts the volume of investment, and its effects change with

institutions and scale. Third, EU accession draws investment into candidate countries, but the evidence is measured at the level of the country. Fourth, subnational research shows that FDI changes the places it reaches and that local politics can steer where it goes, in one case including its type, while ethnicity enters this work only as a network. Fifth, in Bosnia, ethnicity is built into the constitution, structures local politics, and is bound up with the EU path, whereas Croatia completed that path as a single state. None of these literatures asks whether the ethnic composition of a municipality, read as a signal of its orientation toward the EU, predicts the type of FDI it attracts, or whether that relationship changes as a country moves through accession. The argument that follows explains why it should.

3 Theory and Hypotheses

This section develops the argument in four steps. First, it outlines why the two types of FDI respond differently to political risk. Second, it argues that in Bosnia and Croatia, a municipality's ethnic composition carries information about that risk. Third, it connects the two and derives the first hypothesis. Fourth, it treats EU accession as a moderator of that relationship and derives the second. A final subsection sets the scope of the argument for Croatia.

3.1 Two Types of FDI, Two Sensitivities to Risk

The argument starts from the distinction drawn in Section 1. FDI is not one thing, and its two types do not weigh a location in the same way (Pandya 2016). The difference lies in how much choice the investor has.

Market-seeking investment is tied to its markets. A firm that wants to sell to consumers in a country has to produce there. Fixed features of that country, such as market size, largely decide where the investment will go (Pandya 2016, 457). The political environment of a given place matters less directly to the decision. The firm can also limit what it puts at risk. In market-seeking FDI, firms can cope with weak property rights by transferring older production technologies (Pandya 2016, 469).

Export-oriented investment is not bound by these constraints. Its output is sold abroad, so the host market does not direct where production goes. The firm can place it wherever costs are lowest, so locations compete for the same project. Export-oriented production also requires the firm to transfer its most productive, firm-specific assets (Pandya 2016, 469). While the firm has more to lose in adverse conditions, it also has more options in where to go should they catch the turn early. In Pandya's words, export-oriented FDI "imparts more flexibility in location choices such that MNCs can be less tolerant of poor investment climates" (Pandya 2016, 457). Consequently, political risk should deter export-oriented investment more than market-seeking investment (Pandya 2016, 469).

This difference in sensitivity carries two implications for the argument. First, the political environment should shape not only how much FDI a location receives but also what kind. When the risk is higher, export-oriented projects should be scarcer relative to market-seeking ones, even if total investment does not fall. Second, signals that reduce uncertainty about the political environment should matter most to the investors with the most flexibility. Pandya makes this point in relation to national institutions and international agreements that signal credibility (Pandya 2016, 469). This paper applies the same logic below the national level. If something about

a municipality tells an investor about political risk, the export-oriented share is where that information should show.

3.2 Ethnic Composition as a Political Signal

The political risk literature has long treated ethnicity as a source of risk. Busse and Hefeker (2007) include ethnic tensions among the indicators of political risk and institutional quality that shape FDI flows to developing countries. In that framework, ethnic tension is a property of the country that predicts how much investment will arrive. This paper departs from that view in two ways. First, it moves the unit of analysis below the national level, to the municipality. Second, it treats ethnic composition as a directional signal instead of a general risk that deters all investment. In other words, it tells investors how firmly a location is aligned with the EU, which in turn may shape the type of investment it attracts.

The argument rests on what investors can observe. Political risk at the municipal level is hard to measure directly. Ethnic composition is not. It is recorded in the census, changes slowly, and, in Bosnia and Croatia, is closely tied to political orientation (Hadzic, Carlson, and Tavits 2020; Glaurdić, Mochtak, and Lesschaeve 2023). An investor choosing between two municipalities in Bosnia does not need to study local politics to know which group holds the majority. The majority group, in turn, says something about how firmly a municipality is oriented toward the EU.

In Bosnia, this orientation divides along both ethnic and entity lines. Significant segments of public opinion in Republika Srpska openly reject pro-EU policies (Jović 2018). Support for accession is high across the country, but respondents in the Federation are more enthusiastic about it than those in Republika Srpska (Aćić and Grujić 2026). In other words, the difference between the entities is one of degree, not of outright opposition. An investor comparing two locations only needs to know which of the two is more firmly oriented toward the EU.

Most of this evidence, however, describes the entities rather than individual municipalities. In Bosnia, the two largely coincide. Of the 64 Serb-majority municipalities, 60 are in Republika Srpska. Only four Serb-majority municipalities lie in the Federation, and only two Bosniak-majority municipalities lie in Republika Srpska. This paper therefore reads a Bosniak majority as a marker of the EU-aligned side and a Serb majority as a marker of the EU-resistant side. The same overlap has a cost. Ethnic composition and entity cannot be fully separated (Section 4.8).

Croatia is different because EU alignment is a property of the state rather than of public opinion. Croatia signed its SAA in 2001 and joined the EU in 2013. Croat-majority municipalities sit within this trajectory, and Section 3.5 discusses what that means for the Croatian case.

3.3 From Signal to FDI Type

Export-oriented investors are the most sensitive to political risk because they have more to lose and more places to locate. A municipality's ethnic majority tells them how firmly that place is oriented toward the EU. Credibility is the connector between the two. A municipality on the EU-aligned side sits within a community whose public broadly supports the path toward EU regulation, which promises a more predictable environment for an investor. A municipality on the EU-resistant side offers less certainty about where its politics, and by extension its laws, will lean. Signals of credibility weigh most on the investor with the most flexibility (Pandya 2016,

469). While made with national institutions in mind, this paper extends that logic to the municipal level.

Two predictions follow. Export-oriented investors should favor EU-aligned municipalities more strongly than market-seeking investors do. Market-seeking investment should follow demand, largely regardless of the signal. The difference should therefore appear in the composition of investment, not only in its volume. This leads to the first hypothesis:

> The ethnic composition of a municipality systematically predicts the composition of inward FDI, with majority Bosniak municipalities attracting a higher share of export-oriented relative to market-seeking FDI than comparable majority Serb municipalities.

In practice, this hypothesis is tested on the Bosnian case. Section 3.5 explains why Croatia cannot provide the same comparison. Bosnia's 24 Croat-majority municipalities enter the analysis as a separate category, but the first hypothesis makes no prediction for them. The comparison on which the argument rests is between Bosniak- and Serb-majority municipalities.

The same logic has a second implication, which shapes the design but is not stated as a hypothesis. If a weak signal deters export-oriented investors, it may keep some projects away from a municipality altogether rather than only change the mix. The type of FDI can only be observed where FDI arrives. For this reason, the paper models whether investment arrives alongside what kind arrives. Finally, the argument does not claim to be the only route from ethnic composition to FDI type. The same pattern could arise through a different channel. If investors orient toward municipalities whose political orientation matches that of their home country, ethnic composition would predict FDI type through who invests rather than through how investors read risk (see Ledyaeva, Karhunen, and Kosonen 2013). This paper does not separate the two channels.

3.4 Accession as a Moderator

The link between ethnic composition and FDI type, however, may not be static. Neither Bosnia nor Croatia holds the same position toward the EU throughout the period. Each moves through a sequence of formal steps. Croatia signed its SAA in 2001 and became a member in 2013 (Stabilisation and Association Agreement with Croatia 2001; Treaty of Accession of Croatia 2011). Bosnia signed its SAA in 2008, which entered into force in 2015, and received candidate status in December 2022 (Stabilisation and Association Agreement with Bosnia and Herzegovina 2008; European Council 2022). Accession is sequential and externally monitored, and each step depends on reforms. As a country advances, the EU path becomes less a matter of orientation and more a matter of binding commitments. The question is what this does to the signal sent by ethnic composition.

The argument can run in two directions. First, accession could strengthen the signal. As EU rules draw closer, a municipality's orientation toward them carries more weight. Before the path is concrete, the difference between an EU-aligned and an EU-resistant place is largely a difference in attitude. Once the country takes on obligations, the difference becomes whether reforms will be carried out locally or resisted. For an export-oriented investor committing its most productive assets over a long horizon, the gap between the two kinds of place should then matter more.

Second, accession could weaken the signal. The commitments made in the accession process are national. They bind the whole state, not only its EU-aligned parts, and an outside actor that does not depend on local preferences monitors them. International agreements are themselves signals of credibility (Pandya 2016, 469). As this external anchor grows stronger, it may stand in for the local signal and narrow the differences in risk between municipalities. An investor would then have less reason to read the ethnic composition of a place, because the rules it will face are increasingly set elsewhere.

This paper does not choose between the two. Both follow from the same logic, and the theory offers no firm reason to expect one over the other. The second hypothesis therefore states a change in the strength of the relationship and leaves its direction open:

> The strength of this relationship changes as Croatia and Bosnia progress through the EU accession process, as formal institutional alignment increases across regions.

The second hypothesis is a claim about moderation, not a main effect. It does not predict that FDI as a whole rises or changes its mix after a milestone. It predicts that the gap between Bosniak- and Serb-majority municipalities described in the first hypothesis changes in size as accession advances. In Bosnia, that gap can be compared across two milestones within the period. Croatia, which completed accession, offers the contrasting case, within limits that Section 3.5 sets out.

3.5 Scope: The Croatian Case

Croatia was included to give the second hypothesis a completed arc. Unlike Bosnia, it moved through the full sequence within the period, from the years after its SAA to membership in 2013. If accession changes the strength of the ethnic signal, Croatia is where that change should be visible from start to finish. The argument applies to Croatia, however, with two limits.

First, the signal itself is weaker. In Bosnia, the Serb-majority side has an institutional home in Republika Srpska, and the evidence in Section 3.2 describes an entity, not only a population. Croatian Serb-majority municipalities have no such home. They are governed within the same state institutions as the rest of the country, and that state pursued accession. Their majority therefore says less about the political direction of a place than a Serb majority does in Bosnia. These municipalities also lie largely in areas such as Banija, Lika, and parts of Eastern Slavonia, which were among the most affected by the war of 1991–1995. In Banija alone, the population fell by 44.9% between the 1991 and 2001 censuses (Kovjanić et al. 2023). Whatever their ethnic composition signals is difficult to separate from that history.

Second, the comparison cannot be made. Croatia has 18 Serb-majority municipalities under the 2011 census and 16 under the 2021 census, and none of them records FDI in any year (Section 4.4). There is no ethnic gap in FDI type to measure, and therefore no change in that gap across accession.

Consequently, Croatia serves as a benchmark rather than a test. It shows how FDI in EU-aligned, Croat-majority municipalities develops in a country that completed accession, which gives the Bosnian estimates a point of comparison. It does not test either hypothesis the way the Bosnian case does. The argument of this section is thus made for Bosnia, and Croatia marks the limits of where it reaches.

4 Data and Research Design

This section describes how the data was built and how the analysis uses each component. First, it introduces the FDI records and how they are classified, matched to municipalities, and assembled into a panel. Second, it sets out the estimation strategy, the identification of the accession effect, and the main threats to that identification.

4.1 FDI Data

The FDI data comes from fDi Markets, a project-level database of cross-border greenfield investment maintained by the Financial Times (The Financial Times Ltd. 2026).[1] It compiles information from news wires, industry organizations, and media sources (Rommel, Palmtag, and Messerschmidt 2025). This paper starts from every project recorded for Bosnia and Croatia between 2003 and 2023.

Each record is one investment project, and fDi Markets distinguishes three project types. The first is a new project, in which the investor builds a new facility in the host country. The second is an expansion, in which the investor adds capacity to an existing facility. The third is a co-location, in which the investor places a new activity at a site it already operates. All three types are retained, and each counts as one project. Each record names the foreign investor, project type, sector, business activity, destination city, and announcement date. It also reports the project's capital expenditure and expected jobs, both of which fDi Markets partly estimates rather than takes from the investor (Rommel, Palmtag, and Messerschmidt 2025). For this reason, the paper does not use capital expenditure or jobs and is built only on project counts, sectors, and business activities.

Greenfield project data suits this paper's research question for two reasons. First, each project records a location-specific decision. Whatever the project type, an investor must choose where to commit capital. If ethnic composition signals anything to investors, it should register at this point. Aggregate FDI statistics, by contrast, combine various forms of investment and are reported at the national level. Second, each project record carries its sector and business activity, which together allow each project to be classified as export-oriented or market-seeking (Section 4.2).

One limit should be stated plainly. fDi Markets records announcements, not completed investments, so some announced projects may have been scaled back or never built. The data therefore measures investor intent at the moment of the location decision. That is the moment the argument is about, but it is not a measure of capital actually installed.

Between 2003 and 2023, fDi Markets records 492 projects in Bosnia and 934 in Croatia. Of these, 136 in Bosnia and 180 in Croatia list no destination city and carry no coordinates. They cannot be placed in a municipality and are excluded. This leaves 356 projects in Bosnia and 754 in Croatia. Under the classification described in Section 4.2, 121 of Bosnia's projects are export-oriented and 235 market-seeking. In Croatia, the split is 113 and 641.

[1] The data was provided by Tobias Rommel.

4.2 Classifying FDI Type

The distinction at the center of this paper comes from Pandya (2016). Export-oriented investment is production in a host country for sale abroad. It requires the firm to transfer productive assets, which makes it more sensitive to locational conditions. Market-seeking investment, by contrast, supplies the host market, so its location follows local demand. Pandya (2016) calls this type "market-oriented"; this paper uses "market-seeking." fDi Markets does not record the motive of investments. Instead, this paper infers the type from what each record does contain, in two steps.

First, each of the 37 sectors in the data is assigned to one type. The conceptual divide is whether a sector's output can travel. The 24 export-oriented sectors are mainly manufacturing industries whose goods trade across borders, such as Automotive Components and Pharmaceuticals. Investors in these sectors serve foreign markets from the host country, which makes their location choice sensitive to local conditions. The remaining 13 sectors serve the host market directly and are coded as market-seeking. Appendix A lists the full key.

Second, the sector code is checked against the project's business activity. A sector describes what a company makes or provides, and the activity describes what the specific project does. The two can diverge. For example, a clothing retailer and a garment factory share the Textiles sector, yet only the factory produces for export. The store serves local shoppers, which is the definition of market-seeking investment. The paper therefore codes every project whose activity is Retail or Sales, Marketing & Support as market-seeking, whatever its sector. With the sector key alone, 194 of Bosnia's 356 projects would count as export-oriented. With the rule, the number falls to 121. In Croatia, the count falls from 379 to 113, because clothing stores, supermarkets, and drugstores make up a majority of the sector-only export category (Table 1).

Table 1: *FDI projects by type, sector key, and final classification*

	Projects	Export-oriented (sector key)	Export-oriented (final)	Market-seeking (final)	Moved by activity rule
Bosnia	356	194	121	235	73
Croatia	754	379	113	641	266

Located projects, 2003–2023. "Moved" counts projects in export-oriented sectors whose activity is Retail (Bosnia 67, Croatia 248) or Sales, Marketing & Support (Bosnia 6, Croatia 18).

The rule only ever moves projects in one direction. It never turns a market-seeking project into an export-oriented one. This makes the export-oriented category stricter than under the sector key alone. Of the remaining export-oriented projects, 111 of 121 in Bosnia and 75 of 113 in Croatia are manufacturing. Remaining misclassifications add noise to the outcome. As long as they are unrelated to ethnic composition, they do not bias the ethnic estimates but reduce their precision, which makes modest differences harder to detect (Section 5).

Several assignments are judgment calls. First, Software & IT Services is coded as market-seeking. Some software projects serve foreign clients and would count as export-oriented. Most, however, are sales offices or business services (10 of 17 in Bosnia and 26 of 45 in Croatia). Second, Hotels & Tourism is coded as market-seeking, although many of its customers are foreign visitors. The defining property is that tourism is consumed where it is produced and cannot relocate. This judgment affects 31 projects in Croatia and 3 in Bosnia. Third, Food, Beverages & Tobacco is coded as export-oriented, although processed food primarily serves the host market. After the activity rule, 19 of these projects remain export-oriented in Bosnia

and 16 in Croatia. Fourth, Building & Construction Materials is coded as market-seeking, although most of its projects are factories (11 of 12 in Bosnia and all 8 in Croatia), because construction materials rarely travel far from where they are made. Finally, projects in export-oriented sectors whose activity is not manufacturing, mostly logistics, research, and servicing, keep their export-oriented code (10 in Bosnia and 38 in Croatia). Some of these, such as distribution centers, may serve the host market.

In short, the export-oriented category is built around production for foreign markets, and the activity rule makes it stricter than the sector key alone. The classification is an inference from sector and activity, not a direct observation of where a project's output is sold.

4.3 Geography and Ethnic Composition

The unit of analysis is the municipality. Ethnic composition varies sharply within both countries, often between neighboring municipalities. Any coarser unit would blur the variation on which this paper relies. In Bosnia, the two entities (the Federation of Bosnia and Herzegovina and Republika Srpska) and the cantons of the Federation sit above the municipality level, and Brčko District forms a unit of its own. The municipality is also the level at which the census reports ethnic composition and to which each project can be matched. It is therefore the level at which an investor's location choice can be observed.

Each project is placed in a municipality using the coordinates in the geocoded fDi Markets data. These coordinates are matched to municipal boundaries from the GADM database, version 4.1 (GADM 2022), at Level 3 for Bosnia and Level 2 for Croatia. The boundary names are then matched to the names used in the censuses. An alias list resolves differences in spelling between the two sources, for example bilingual Croatian–Italian names in Istria, such as Pula–Pola. One Croatian project fell in a GADM unit that does not exist in the census. It was assigned by hand to Solin, the city listed on its record. After these corrections, every located project is assigned to a municipality, for a total of 356 projects in Bosnia and 754 in Croatia.

Separate handling is required for two features of the Croatian data. First, the census reports the City of Zagreb by its 17 city districts, while fDi Markets records Zagreb projects only at the city level. This paper therefore treats Zagreb as a single unit, using the census figures for the city as a whole. Second, the names Otok, Privlaka, and Sveta Nedelja each belong to two municipalities in different counties, which are kept apart by county. In total, the Croatian panel contains 556 units, made up of 555 towns and municipalities and the City of Zagreb.

Ethnic composition comes from the national censuses. For Bosnia, this paper uses the 2013 census for all 142 municipalities in the panel (Agency for Statistics of Bosnia and Herzegovina 2016). It is the only Bosnian census conducted since the end of the Yugoslav Wars. For Croatia, the 2011 and 2021 censuses are combined (Croatian Bureau of Statistics 2012, 2022). The 2011 census is used for 2003–2016 and the 2021 census for 2017–2023. For every municipality, the shares of Bosniaks, Croats, and Serbs are recorded. The majority group is the group with the largest share. In this analysis, 54 Bosnian municipalities are Bosniak-majority, 24 Croat-majority, and 64 Serb-majority. Croatia has 18 Serb-majority municipalities under the 2011 census and 16 under the 2021 census.

The Bosnian census carries two limits. First, because this paper uses a single census, ethnic composition is assumed fixed for two decades, from 2003 through 2023. Composition will have changed over this period, for example through postwar returns. Second, the census itself is

contested. The three statistical agencies that conducted it broke down in disagreement, and the census produced two competing official results, each valid in a different part of the country (Hayden 2021). Comparing the census with voting patterns, Žíla and Čermák (2021) find discrepancies that call into question the official numbers of Bosniak and Croat returnees in Republika Srpska. Both limits mean that the ethnic shares are measured with error. If that error is unrelated to FDI, it biases the ethnic coefficients toward zero, which works against finding an ethnic effect. The disputed counts, however, concentrate in particular locations, so the error is not purely random (Hayden 2021; Žíla and Čermák 2021). The categorical majority measure is less exposed to it than the continuous share, because a municipality's largest group rarely changes when a minority is overcounted.

4.4 The Panel

The located projects are aggregated into a municipality-year panel covering 2003 to 2023. Every municipality appears in every year, including years without FDI. For each municipality-year, the panel records the number of projects, how many are export-oriented, and how many are market-seeking. It also records the ethnic composition described in Section 4.3. The Bosnian panel contains 142 municipalities over 21 years, for a total of 2,982 municipality-years. The Croatian panel contains 556 units over the same period, for a total of 11,676 municipality-years (Table 2).

Table 2: *The municipality-year panels*

	Units	Years	Municipality-years	Active (share)	Ever active	Projects	Export-oriented	Market-seeking
Bosnia	142	2003–2023	2,982	201 (6.7%)	55	356	121	235
Croatia	556	2003–2023	11,676	293 (2.5%)	86	754	113	641

Active = municipality-years with at least one project; ever active = units with at least one active year. Croatia counts the City of Zagreb as one unit. Bosnian units by entity: Federation 79, Republika Srpska 62, Brčko District 1.

Retaining zero entries was a deliberate choice. A panel built only from years with FDI would describe what type of investment arrives but leave out whether any arrives at all. It would also condition the comparison on an outcome that may itself depend on ethnic composition. The zeros carry information and are modeled in Section 4.6.

FDI in both countries is rare and concentrated. Only 201 of Bosnia's 2,982 municipality-years, or 6.7%, receive any project, and 55 of its 142 municipalities receive FDI at least once. In Croatia, 293 municipality-years, or 2.5%, receive any project. A few cities account for most projects. Novo Sarajevo, Banja Luka, Grad Mostar, Tuzla, and Zenica hold about 58% of Bosnian projects. In Croatia, Zagreb alone accounts for 37%, and together with Split, Rijeka, Osijek, and Zadar for 60%.

Where FDI is observed, the export-oriented share is usually at one extreme. A majority of active municipality-years receive exactly one project, making their share either 0 or 1. In Bosnia, 94 of the 201 active municipality-years are entirely market-seeking and 83 entirely export-oriented. In Croatia, 196 of 293 active municipality-years are entirely market-seeking and 57 entirely export-oriented. Of the 62 municipality-years in Bosnia and 94 in Croatia with more than one project, only 24 and 40 take an intermediate value.

The two panels are kept separate rather than pooled. Each country passes through different accession milestones, and Croatia offers no internal ethnic contrast. Its Serb-majority

municipalities receive no FDI in any year. Bosnia therefore carries the analysis, and Croatia serves as the benchmark of a country that completed accession.

4.5 Variables

The analysis uses two outcomes, one for each margin of investor choice. The first, has_fdi, equals 1 if a municipality records at least one greenfield project in a given year and 0 if not. It captures the extensive margin, which is whether FDI arrives at all. The second, export_share, is the number of export-oriented projects divided by all projects a municipality records in that year. It captures the intensive margin, which is what kind of FDI arrives, given that some does. Because export_share is defined only in active years, Bosnia is left with 201 municipality-years and Croatia with 293.

Ethnic composition also enters in two forms. The continuous measure is the Bosniak share of the municipal population. The categorical measure is the majority group defined in Section 4.3 (Bosniak-, Croat-, or Serb-majority). Its coefficients show how Bosniak- and Croat-majority municipalities differ from comparable Serb-majority municipalities, holding the other variables fixed. The two measures ask slightly different questions. The share asks whether a larger Bosniak population leans toward more export-oriented FDI. The majority measure asks whether the identity of the dominant group does the leaning. As noted in Section 4.3, the categorical measure is also less exposed to errors in the census shares. In Croatia, where Croats are the largest group in all but 18 municipalities under the 2011 census (16 under the 2021 census), the models use the continuous Serb share instead.

Accession enters as period indicators. For Bosnia, post_saa equals 1 from 2008, the year Bosnia signed the SAA, and post_candidacy equals 1 from 2022, the year candidate status was granted. The two are nested and divide the 21 years into three periods, Pre-SAA (2003–2007), Post-SAA (2008–2021), and Post-candidacy (2022–2023). The last period spans just two years. For Croatia, post_accession equals 1 from 2013 as the year of EU membership; as with candidacy, the whole year counts, although Croatia joined on July 1. This splits the panel into pre- and post-accession periods.

All models control for the log of municipal population, taken from the same census as the ethnic shares. Population matters because FDI concentrates in large municipalities, and Bosniak-majority municipalities are larger on average. The median Bosniak-majority municipality has about 25,000 residents, the median Serb-majority municipality about 11,000. Without this control, the ethnic coefficient would partly pick up size. In Bosnia, population and ethnic composition are constant over time within a municipality, because the single 2013 census is used for all years. In Croatia, both change in 2017, when the panel switches from the 2011 to the 2021 census. Finally, each Bosnian municipality is assigned to its entity. The Federation has 79 municipalities, Republika Srpska 62, and Brčko District one. These assignments define the entity fixed effects discussed in Section 4.6. Table 3 lists each variable with its coding and source.

Table 3: *Variables, coding, and sources*

Variable	Coding	Used in	Source
has_fdi	1 if the municipality records at least one project in the year; 0 otherwise	All municipality-years (Bosnia 2,982; Croatia 11,676)	fDi Markets
export_share	Export-oriented projects ÷ all projects in the year	Active years only (Bosnia 201; Croatia 293)	fDi Markets; sector key and activity rule
Bosniak share	Bosniak % of the municipal population	Bosnia	Census
Majority group	Largest of the Bosniak, Croat, and Serb shares; Serb-majority = reference	Bosnia (54 Bosniak, 24 Croat, 64 Serb)	Census
Serb share	Serb % of the municipal population	Croatia	Census
post_saa	1 from 2008 (SAA signed); 0 before	Bosnia	SAA
post_candidacy	1 from 2022 (candidate status); 0 before	Bosnia	European Council conclusions
post_accession	1 from 2013 (EU membership); 0 before	Croatia	Accession treaty
log population	Natural log of the municipal population	Both	Census
Entity	Federation (79), Republika Srpska (62), Brčko District (1)	Bosnia (entity fixed effects)	Census

Census = Bosnia 2013 (Agency for Statistics of Bosnia and Herzegovina 2016), held fixed for 2003–2023; Croatia 2011 for 2003–2016 and 2021 for 2017–2023 (Croatian Bureau of Statistics 2012, 2022). fDi Markets = The Financial Times Ltd. (2026). SAA = Stabilisation and Association Agreement with Bosnia and Herzegovina (2008). European Council conclusions = European Council (2022). Accession treaty = Treaty of Accession of Croatia (2011). Active = municipality-years with at least one project. Bosnian accession periods: Pre-SAA 2003–2007, Post-SAA 2008–2021, Post-candidacy 2022–2023.

4.6 Estimation: A Hurdle (Two-Part) Model

Two features of the data shape the estimation. First, FDI is rare. Only 6.7% of Bosnian and 2.5% of Croatian municipality-years record a project. Second, export share exists only where FDI exists. A municipality without a project does not have a measurable share. A single model of export share would either drop the years without FDI or treat them as market-seeking. Both choices would mix two different questions, whether FDI arrives and what kind arrives. The paper therefore separates them in a hurdle, or two-part, model, which treats the decision to cross zero and the outcome beyond zero as separate processes, each with its own coefficients (Cragg 1971).

The first part models arrival. It is a logit of has_fdi on all municipality-years:

$$\Pr(\text{has_fdi}_{it} = 1) = \Lambda\big(\alpha_1 + \beta_1 \,\text{Ethnic}_i + \gamma_1 \,\log \text{Pop}_i + \delta_{1,p(t)} + \mu_{1,e(i)}\big) \quad (1)$$

Here Λ is the logistic function, Ethnic_i is either the Bosniak share or the set of Bosniak- and Croat-majority indicators, $\delta_{p(t)}$ is the effect of the accession period that year t falls in, and $\mu_{e(i)}$ is the fixed effect of the entity that municipality i belongs to.

The second part models type. It is an ordinary least squares (OLS) regression of export_share on active municipality-years only:

$$\text{export_share}_{it} = \alpha_2 + \beta_2 \,\text{Ethnic}_i + \gamma_2 \,\log \text{Pop}_i + \delta_{2,p(t)} + \mu_{2,e(i)} + \varepsilon_{it}, \quad \text{for has_fdi}_{it} = 1 \quad (2)$$

As Section 4.4 showed, export share is not a smooth outcome but bunches at the extremes, with 177 of Bosnia's 201 active years and 253 of Croatia's 293 taking a value of exactly 0 or 1. OLS on such an outcome is a linear approximation, much as a linear probability model approximates a binary outcome, and fitted values can fall outside the 0–1 range. In exchange, each coefficient reads directly as a change in the export-oriented share, and linear estimates

of this kind typically come close to the marginal effects from nonlinear models (Angrist and Pischke 2009, sec. 3.4.2). The cost is fit at the boundaries, not a systematic tilt in the coefficients.

Both parts share the same structure. First, each model is estimated with both forms of ethnic composition. Second, each is estimated without and with entity fixed effects. Municipality fixed effects are not an option, because ethnic composition in Bosnia does not change within a municipality over time and would be absorbed entirely. Entity fixed effects remove level differences between the Federation and Republika Srpska. The ethnic comparison then rests on variation within entities, which Section 4.8 discusses further. Third, time enters as the three accession periods rather than as 21 year dummies, because FDI events are too few for year effects. The median year has 10 active municipalities in Bosnia, and two years have five or fewer. The three periods, by contrast, hold 53, 126, and 22 active municipality-years. Finally, standard errors are clustered by municipality, because each municipality is observed in 21 years and these years are not independent observations.

The type model in Equation 2 has one important limit. Because it includes only municipality-years in which FDI arrived, it takes arrival as given, although arrival is itself an outcome that ethnic composition may shape. If it does, the active municipality-years form a selected group. The ethnic coefficient in Equation 2 would mix an effect on type with differences in which municipalities receive FDI at all (Angrist and Pischke 2009, sec. 3.4.2). Because the direction of this selection cannot be signed in advance, the type model describes the composition of FDI where it arrives. It is not a clean causal estimate of ethnicity's effect on investment type. This is why the paper reports the two parts side by side rather than relying on the type model alone.

Croatia is estimated with the same two parts in a simpler form. Ethnic composition enters as the continuous Serb share, log population changes once with the switch to the 2021 census, and the period effect separates the years before and after accession in 2013. Croatia has no entities, so there are no entity fixed effects. The arrival model uses all 11,676 municipality-years, and the type model only the 293 active ones. Section 4.7 extends both parts with interactions between ethnic composition and the accession milestones.

4.7 Identification: Accession Moderation as Difference-in-Differences

The second half of the research question asks whether the relationship between ethnic composition and FDI changes as Bosnia moves through the accession process. This is a question of moderation. When the moderating variable is a milestone that switches on in a given year, a moderation test takes the form of a difference-in-differences (DiD). The first difference is between groups, such as Bosniak- against Serb-majority municipalities or a higher against a lower Bosniak share. The second is over time, comparing years before a milestone with years after it. The DiD estimate is the change in the group gap from before to after. The question thus becomes not whether Bosniak-majority municipalities attract more export-oriented FDI, but whether their advantage or disadvantage grows or shrinks once a milestone is passed.

The model extends Equations 1 and 2 by interacting ethnic composition with the two Bosnian milestones:

$$Y_{it} = \alpha + \beta\,\mathrm{Ethnic}_i + \theta_1(\mathrm{Ethnic}_i \times \mathrm{PostSAA}_t) + \theta_2(\mathrm{Ethnic}_i \times \mathrm{PostCand}_t) + \lambda_1\,\mathrm{PostSAA}_t + \lambda_2\,\mathrm{PostCand}_t + \gamma\,\log \mathrm{Pop}_i + \mu_{e(i)} + \varepsilon_{it} \quad (3)$$

Here Y_{it} is either arrival or export share, and the other terms are defined as previously. The two milestone indicators replace the period effects found in Section 4.6. With the indicators nested, λ_1 and λ_2 describe the same three periods. The coefficient β is the ethnic gap before the SAA. The coefficient θ_1 is the change in that gap from the Pre-SAA to the Post-SAA period. The coefficient θ_2 is the additional change once candidacy is granted, on top of the change at the SAA. The ethnic gap in each period therefore builds up step by step. Before the SAA it is simply β. The SAA adds θ_1, so the gap after the SAA is $\beta + \theta_1$. Candidacy adds θ_2 on top, so the gap after candidacy is $\beta + \theta_1 + \theta_2$. The paper reports each of these gaps, with standard errors, alongside the interaction terms. If accession does not moderate the ethnic relationship, both interactions are zero. A joint Wald test checks this directly. For the categorical measure, a second joint test uses only the Bosniak interactions. Croat-majority cells are thin, with only two Croat-majority municipality-years recording FDI after candidacy. The second test keeps them from driving the result.

Every municipality passes each milestone in the same year, so the design has common timing. There are no early- and late-treated groups to compare. Ethnic composition, not the timing of the milestone, defines who is "treated." For consistency, the structure of Section 4.6 is kept. Equation 3 is estimated for both outcomes, with both forms of ethnic composition, with and without entity fixed effects, and with standard errors clustered by municipality.

One adjustment is applied to the arrival model. In a logit, Equation 3 sits inside the logistic function, as in Equation 1. The interaction coefficients are then changes in log-odds, not in probabilities. In nonlinear models of this kind, the effect of an interaction on the probability does not equal the interaction coefficient. It can even take the opposite sign, and standard software output does not test it correctly (Ai and Norton 2003). For this reason, the arrival DiD is also estimated as a linear probability model by OLS on has_fdi. Its interaction terms read directly as percentage-point changes in the yearly probability of receiving FDI. Both versions are reported, so each arrival result appears in eight specifications (two functional forms, two ethnic measures, and two fixed-effect choices). The type model needs no such adjustment because it is already linear.

The Croatian case does not support a DiD of this kind. Its Serb-majority municipalities record no FDI before or after accession, so there is no ethnic gap whose change could be measured. The Croatian estimates therefore serve as a benchmark rather than a test. First, the paper estimates how arrival and export share in Croat-majority municipalities changed after accession in 2013. This shows the trajectory of EU-aligned municipalities in a country that completed accession. Second, for the sake of symmetry with Bosnia, it interacts the continuous Serb share with post_accession. This term rests mainly on Serb minorities inside Croat-majority municipalities. It is not a comparison with Serb-majority municipalities.

A DiD identifies the moderating effect of accession only if the ethnic groups would have followed parallel paths in the absence of the milestone. This assumption cannot be proven. Section 4.8 describes how the paper addresses it, along with the other main threats to identification.

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4.8 Threats to Identification

The design rests on several assumptions that the data cannot fully verify. Some can be tested, some can only be reduced, and some remain as limits on the results. This section takes each in turn.

First, the DiD requires parallel trends. Absent each milestone, the ethnic groups would have had to follow the same path in arrival and export share. This cannot be proven, because a path that did not happen cannot be observed. The paper addresses this threat in two steps. First, it describes how each group's arrival and export share developed before each milestone. For the SAA, this is shown year by year from 2003 to 2007. For candidacy, the years 2008 to 2021 are grouped into four blocks. Second, it tests whether the groups were already moving apart before each milestone. To do so, it adds a yearly time trend for each ethnic group, using only the years before the milestone. If the trends differ significantly, the groups were already drifting apart before the milestone. These tests have limited power. The Pre-SAA period covers only five years and 53 active municipality-years. Of those, Serb-majority municipalities account for only 12. A null pre-trend test is therefore only weak evidence of parallel trends, not proof.

Second, in Bosnia, ethnic composition and entity largely coincide. Of the 64 Serb-majority municipalities in Bosnia, 60 are in Republika Srpska. Only six municipalities cross the entity line (four Serb-majority municipalities in the Federation and two Bosniak-majority municipalities in Republika Srpska). Entity fixed effects remove level differences between the entities. They do not, however, rule out that the entities drift apart after a milestone, for example because of entity-level policy. A shift such as this would look almost identical to an ethnic DiD. The paper therefore adds interactions between Republika Srpska and each milestone, and asks whether the ethnic interaction survives once an entity-level DiD is allowed. The standard errors in this comparison are wide because it rests on the six crossover municipalities, so the comparison cannot cleanly separate ethnicity from entity.

Third, ethnic composition overlaps with city size. As Section 4.5 showed, Bosniak-majority municipalities are larger on average, and FDI concentrates in large municipalities. The log-population control addresses this in every model. The control captures differences in size across municipalities rather than changes in size over time because population is held fixed at the census value (in Croatia, it changes once, in 2017).

Fourth, the sample of FDI events is small. Bosnia has 201 active municipality-years out of 2,982, and the DiD thins the cells further by group and period. For example, only two Croat-majority municipality-years record FDI after candidacy. The type model is affected the most because it uses only active years. The design can therefore detect large changes in the ethnic gap, but a null result does not rule out modest ones. Low power mostly works against finding an effect, but very thin cells can also produce a significant coefficient on a handful of observations (Section 5.3).

Fifth, a single municipality or an arbitrary date could drive a result. Any significant interaction must therefore pass three checks (Section 5.4). A jackknife re-estimates the model 142 times, dropping one municipality each time, to see whether any single place drives the estimate. A second check drops the five largest FDI destinations (Novo Sarajevo, Banja Luka, Grad Mostar, Tuzla, and Zenica). Third, a placebo test, run only on the Post-SAA years (2008–2021), assigns a false milestone in 2015 and asks whether a shift of similar size appears. One caveat is that 2015 is the year the SAA entered into force, so it is not an event-free date. The placebo

tests whether the post-SAA shift is a gradual drift between 2008 and 2021. It does not test a date on which nothing occurred.

Sixth, the candidacy milestone is coded conservatively. Bosnia was granted candidate status in December 2022, but post_candidacy equals 1 for the whole of 2022. Most of that year preceded candidacy, yet it counts as post-candidacy. This biases the candidacy interaction toward zero. Combined with the short post-candidacy period of only two years, it makes the candidacy test the weakest in the design.

Seventh, Croatia exhibits no internal ethnic contrast. Its Serb-majority municipalities record no FDI in any year. As Section 4.7 explained, the Croatian estimates serve as a benchmark for a completed accession rather than a proper ethnic DiD test.

Finally, the measures of ethnic composition come from the census, so they share the census problems described in Section 4.3. Bosnia's 2013 census is used for all 21 years, and the counts are disputed. Both problems add error to the ethnic shares. If that error is unrelated to FDI, it pushes the ethnic coefficients toward zero and makes an ethnic effect harder to find.

In summary, the design can show whether the ethnic gap in arrival and export share changed when Bosnia passed each milestone, and whether that change survives the checks detailed above. However, it cannot fully separate an ethnic shift from an entity-level shift. Nor can it rule out modest effects that the small sample cannot detect, or establish parallel trends beyond the evidence the pre-period provides.

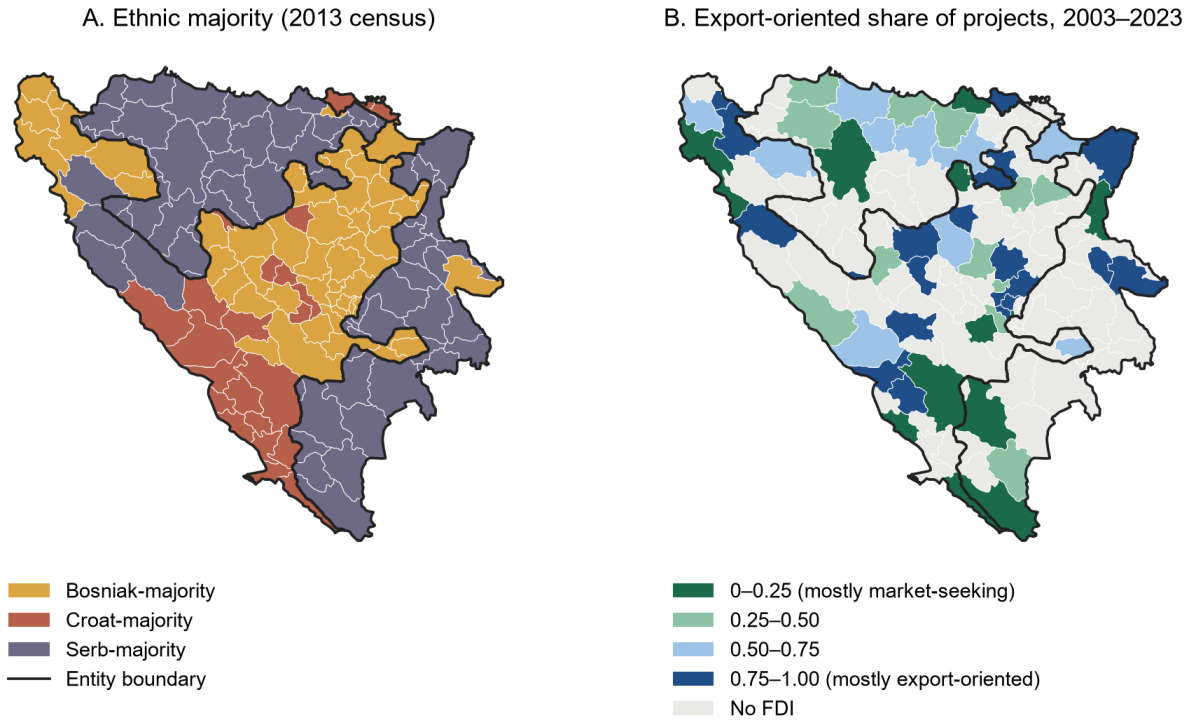
5 Results

This section presents the results in five steps. First, it describes how participation and the export-oriented share differ across ethnic groups and accession periods in Bosnia. Second, it reports the two-part model, which asks whether ethnic composition predicts whether FDI arrives and, where it does, what kind arrives. This is the test of the first hypothesis. Third, it asks whether the ethnic gap changes after the SAA and after candidacy, which tests the second hypothesis. Fourth, it puts the one significant interaction through the checks set out in Section 4.8. Fifth, it reports the Croatian benchmark.

5.1 Descriptive Patterns

Figure 1 sets two maps side by side. Panel A shows the ethnic majority of each municipality. Panel B shows the export-oriented share of all projects a municipality received between 2003 and 2023. Two features stand out. First, ethnic composition and entity nearly coincide. Serb-majority municipalities make up almost all of Republika Srpska, and Bosniak- and Croat-majority municipalities make up almost all of the Federation. Second, Panel B does not reproduce that line. Only 55 of the 142 municipalities receive any project in the period (26 Bosniak-majority, 19 Serb-majority, and 10 Croat-majority). Municipalities with mostly export-oriented and mostly market-seeking FDI appear on both sides of the entity boundary. The three largest destinations, one from each group, all receive mostly market-seeking FDI: Novo Sarajevo (5 of 102 projects export-oriented), Banja Luka (13 of 52), and Grad Mostar (1 of 20).

Figure 1: *Ethnic majority and FDI type across Bosnian municipalities*



Panel A: the largest of the Bosniak, Croat, and Serb shares, 2013 census. Panel B: export-oriented projects ÷ all projects per municipality, 2003–2023; classes include their upper bound; gray = no project in the period. Usora (Croat-majority, no FDI) is drawn within Tešanj because the GADM boundaries do not separate the two. Boundaries: GADM 4.1, level 3 (GADM 2022).

Table 4 reports the same comparison over time. Across all years, the raw data runs against the first hypothesis. Serb-majority municipalities show the highest export-oriented share, 0.423 of their projects, against 0.307 in Bosniak-majority and 0.273 in Croat-majority municipalities. The equal-weighted mean across active years puts Serb- and Bosniak-majority municipalities in the same order (0.506 and 0.466). The two measures differ mainly because of Novo Sarajevo. It holds 102 of the 212 Bosniak-majority projects, and only five of them are export-oriented. Most FDI in Bosnia is market-seeking (235 of 356 projects, or 66%).

The clearer difference lies in participation. Bosniak-majority municipalities receive FDI in 9.6% of municipality-years, against 5.1% in Serb-majority and 4.8% in Croat-majority municipalities. They also receive almost twice as many projects as Serb-majority municipalities (212 against 111). These figures are unadjusted. Section 5.2 shows what remains of them once municipal size is held constant.

Split by accession period, both patterns change. Before the SAA, Bosniak-majority municipalities held an edge on both margins. Their export-oriented share was 0.362 against 0.286 in Serb-majority municipalities, and they received FDI in 13.0% of municipality-years against 3.8%. After 2008, the export-oriented share fell to 0.266 in Bosniak-majority municipalities and rose to 0.429 in Serb-majority ones. The Bosniak–Serb gap therefore reversed, from +0.077 to −0.162. After candidacy, it reached −0.247, on 19 and 13 projects. Participation, in turn, converged. It fell to 8.2% in Bosniak-majority municipalities and rose to 5.4% in Serb-majority ones. For the country as a whole, the export-oriented share was flat from the Pre-SAA to the Post-SAA period (0.330 and 0.320) and rose only in the two candidacy years (0.500, on 34 projects).

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Several cells rest on very few projects. The Croat-majority share of 1.000 after candidacy reflects two projects, and the Pre-SAA figure of 0.143 reflects seven. The Pre-SAA comparison as a whole rests on 53 active municipality-years, of which 12 are Serb-majority. These figures describe the data. They do not control for size and carry no inference. The next subsection tests them.

Table 4: *Participation and export-oriented share by majority group and accession period, Bosnia*

Group	Period	Municipality-years	Active	Participation	Projects	Export-oriented	Pooled share	Mean share (active)
Bosniak	Pre-SAA	270	35	0.130	69	25	0.362	0.529
	Post-SAA	756	62	0.082	124	33	0.266	0.421
	Post-candidacy	108	12	0.111	19	7	0.368	0.514
	All years	1,134	109	0.096	212	65	0.307	0.466
Croat	Pre-SAA	120	6	0.050	7	1	0.143	0.167
	Post-SAA	336	16	0.048	24	6	0.250	0.344
	Post-candidacy	48	2	0.042	2	2	1.000	1.000
	All years	504	24	0.048	33	9	0.273	0.354
Serb	Pre-SAA	320	12	0.038	21	6	0.286	0.325
	Post-SAA	896	48	0.054	77	33	0.429	0.528
	Post-candidacy	128	8	0.063	13	8	0.615	0.646
	All years	1,344	68	0.051	111	47	0.423	0.506
All	Pre-SAA	710	53	0.075	97	32	0.330	0.442
	Post-SAA	1,988	126	0.063	225	72	0.320	0.452
	Post-candidacy	284	22	0.077	34	17	0.500	0.606
	All years	2,982	201	0.067	356	121	0.340	0.466

Participation = share of municipality-years with at least one project. Pooled share = export-oriented projects ÷ all projects (project-weighted). Mean share = average export_share across active municipality-years (equal-weighted). Majority group from the 2013 census (Agency for Statistics of Bosnia and Herzegovina 2016). Pre-SAA 2003–2007, Post-SAA 2008–2021, Post-candidacy 2022–2023. Croat Post-candidacy rests on two projects and Croat Pre-SAA on seven.

5.2 Ethnic Composition, Arrival, and Type

Table 5 reports the two-part model. Columns 1–4 model arrival on all 2,982 municipality-years, and columns 5–8 model the export-oriented share on the 201 active ones. Each part is estimated with the continuous Bosniak share and with the majority-group indicators, without and with entity fixed effects.

Ethnic composition does not predict whether FDI arrives once municipal size is held constant. The Bosniak share is −0.002 ($p = .695$), and it stays at −0.002 ($p = .684$) with entity fixed effects. Without entity fixed effects, the categorical measure is null as well. Bosniak- and Croat-majority municipalities are no more likely to receive FDI than Serb-majority ones (−0.023, $p = .949$; −0.123, $p = .717$). What predicts arrival is size. Log population enters at between +1.562 and +1.611 in all four columns ($p < .001$). The participation gap in Section 5.1 therefore does not survive this control. Bosniak-majority municipalities receive FDI more often, but the difference reflects their larger size, not their ethnic composition.

One specification departs from this pattern. With entity fixed effects, the categorical measure returns +1.798 for Bosniak-majority and +1.719 for Croat-majority municipalities ($p = .019$ and $p = .047$). This is an artifact, not a finding, for three reasons. First, it contradicts the continuous measure in the same specification, which stays null. Second, the two coefficients are nearly identical, so the model separates Serb-majority municipalities from all others rather than

Bosniak- from Serb-majority ones. Third, with entity fixed effects, the comparison rests on the six municipalities that cross the entity line. The four Serb-majority municipalities in the Federation record FDI in a single municipality-year over the entire period. Within the Federation, the contrast is therefore between Serb-majority places that almost never receive FDI and every other municipality.

On type, the first hypothesis is not supported. Among municipality-years with FDI, Bosniak-majority municipalities show no higher export-oriented share than Serb-majority ones. The coefficient is −0.058 (p = .607) without entity fixed effects and +0.200 (p = .256) with them. The continuous measure is null as well (−0.001, p = .590; +0.002, p = .350). No ethnic coefficient in columns 5–8 reaches p < .10. The lowest, Croat- against Serb-majority in column 7, has p = .188. Section 3.3 makes no prediction for Croat-majority municipalities, so that coefficient is reported but not interpreted.

Size enters the type model with the opposite sign. Log population lies between −0.127 and −0.134 in all four columns (p ≤ .001). Larger municipalities receive FDI more often, and the FDI they receive is less often export-oriented. Section 6 returns to this pattern.

These estimates have two limits. First, the type model rests on 201 observations, and classification noise in the outcome reduces precision further (Section 4.2). Without entity fixed effects, the 95% confidence interval for the Bosniak–Serb coefficient runs from −0.280 to +0.163. It excludes a Bosniak-majority advantage larger than about 0.16 in the export-oriented share. With entity fixed effects, the interval widens to −0.145 to +0.545. A modest effect therefore cannot be ruled out. Second, the type model conditions on arrival (Section 4.6). It describes the composition of FDI where FDI arrives. It is not a clean estimate of what ethnic composition does to the type of investment a municipality attracts. In short, ethnic composition predicts neither whether FDI arrives nor what kind arrives, once size is held constant. The first hypothesis is not supported.

Table 5: *Two-part model of FDI arrival and type, Bosnia*

	Arrival (logit)				Type (OLS)			
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Bosniak share	−0.002	−0.002			−0.001	+0.002		
	(0.005)	(0.005)			(0.001)	(0.002)		
Bosniak-majority			−0.023	+1.798**			−0.058	+0.200
			(0.362)	(0.766)			(0.113)	(0.176)
Croat-majority			−0.123	+1.719**			−0.169	+0.113
			(0.340)	(0.866)			(0.129)	(0.211)
Log population	+1.581***	+1.599***	+1.562***	+1.611***	−0.127***	−0.134***	−0.130***	−0.129***
	(0.211)	(0.224)	(0.215)	(0.222)	(0.040)	(0.037)	(0.038)	(0.038)
Post-SAA	−0.208	−0.208	−0.208	−0.209	−0.009	+0.006	−0.009	+0.011
	(0.154)	(0.154)	(0.153)	(0.155)	(0.075)	(0.075)	(0.077)	(0.076)
Post-candidacy	+0.048	+0.048	+0.048	+0.048	+0.130	+0.144	+0.124	+0.152
	(0.314)	(0.315)	(0.314)	(0.315)	(0.108)	(0.108)	(0.108)	(0.108)
Entity FE	No	Yes	No	Yes	No	Yes	No	Yes
N	2,982	2,982	2,982	2,982	201	201	201	201
Pseudo R^2 / R^2	0.225	0.226	0.225	0.230	0.073	0.101	0.082	0.103

Standard errors in parentheses, clustered by municipality. Columns 1–4: logit coefficients (log-odds) on all municipality-years. Columns 5–8: OLS on active municipality-years, with export_share as the outcome. Bosniak share is in percentage points. Reference categories: Serb-majority and Pre-SAA. Fit is McFadden's pseudo R^2 in columns 1–4 and R^2 in columns 5–8. *** p < .01, ** p < .05, * p < .10.

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5.3 Accession and the Ethnic Gap

This subsection asks whether the relationship between ethnic composition and FDI changes as Bosnia moves through accession. It first examines the trends before each milestone, on which the difference-in-differences rests. It then reports the type DiD, which tests the second hypothesis, followed by the arrival DiD.

Tables B1 and B2 in Appendix B report the evidence on parallel trends. Table B1 shows arrival and the export-oriented share by group, year by year for 2003–2007 and in four blocks for 2008–2021. Table B2 adds a linear time trend for each group within each pre-milestone window. The Bosniak–Serb contrast passes this test in both windows. Before the SAA, the trend interaction is +0.053 for arrival ($p = .799$) and +0.097 for type ($p = .362$). Before candidacy, it is −0.053 for arrival ($p = .345$) and +0.002 for type ($p = .941$). The continuous measure gives the same result (Table B2). The descriptives agree. Between 2003 and 2007, Bosniak-majority municipalities received FDI in 9.3% to 16.7% of municipality-years and Serb-majority municipalities in 3.1% to 6.3%, with no sign that the two were moving apart.

The Croat–Serb contrast does not pass. Croat-majority arrival was already rising relative to Serb-majority arrival before both milestones (+0.546, $p = .048$; +0.155, $p = .014$). The Croat interactions in the arrival DiD therefore cannot be read as DiD estimates. The Bosniak–Serb result has limits as well. The Pre-SAA window covers five years and 53 active municipality-years, of which 12 are Serb-majority, and the type test before the SAA rests on those 53 observations. A null pre-trend is therefore weak evidence of parallel trends, not proof (Section 4.8).

Table 6 reports the type DiD. Accession does not change the ethnic gap in the export-oriented share. None of the Bosniak interactions is significant. The Bosniak–Serb interaction with post-SAA is −0.204 ($p = .212$), and with post-candidacy −0.143 ($p = .414$). The continuous interactions are jointly null, at $p = .327$ without and $p = .325$ with entity fixed effects. For the categorical measure, the Bosniak interactions alone are jointly null as well ($p = .266$ and $p = .372$). The implied gap in each period tells the same story (Table B3). It is +0.117 before the SAA ($p = .456$), −0.087 after it ($p = .480$), and −0.231 after candidacy ($p = .242$). The direction matches the descriptive reversal in Section 5.1, but no period gap can be distinguished from zero. Accession did not make Bosniak-majority municipalities attract a more export-oriented mix of FDI, nor a less export-oriented one. The second hypothesis is not supported.

Two coefficients in Table 6 reach some level of significance, and neither carries weight. First, the Croat-majority interaction with post-candidacy is +0.388 ($p = .025$), or +0.393 ($p = .023$) with entity fixed effects. It is fit to two observations, the only Croat-majority municipality-years with FDI in 2022–2023, both with a share of 1.0. It also makes the joint test of all categorical interactions significant ($p = .004$ and $p = .009$). Second, with entity fixed effects, the Pre-SAA Bosniak–Serb gap is +0.334 ($p = .098$). No Serb-majority municipality in the Federation received FDI before 2008, so within the Federation this gap has no Serb-majority comparison behind it.

Table 7 reports the arrival DiD. Here the ethnic gap does change. The interaction with post-SAA is negative and significant in all eight specifications. In the logit, it is −0.012 per point of Bosniak share ($p = .006$) and −1.046 for Bosniak- against Serb-majority municipalities ($p = .003$). Entity fixed effects leave it unchanged (−0.012 and −1.051). The linear probability model gives the same sign and significance. After 2008, the Bosniak–Serb difference in the yearly

probability of receiving FDI is 6.4 percentage points smaller than before (p = .001). In continuous terms, each additional 10 points of Bosniak share goes with an arrival probability about 0.7 percentage points lower after 2008, relative to before. Joint tests point the same way (Table 7). The test of all interactions is significant except in the categorical logit, where the Croat terms add noise (p = .052 and .055), and the Bosniak interactions alone are jointly significant in every specification (p ≤ .013). The candidacy interaction is null throughout (lowest p = .407).

Panel B of Table 7 turns the interactions back into the gap in each period. The change after 2008 does not open a gap against Bosniak-majority municipalities. Instead, a modest advantage they held before 2008 disappears. In the categorical logit, the Pre-SAA gap is +0.750 (p = .096). After 2008, it is −0.296 (p = .443), and after candidacy −0.126 (p = .836). On its own, the Pre-SAA advantage is only marginal, and in the LPM it is not significant (+0.040, p = .264). What is robust is the change, not either level. Table B1 shows how the change unfolds. In the raw data, the Bosniak–Serb difference in participation narrows in 2008–2011 (7.9% against 4.7%), widens again in 2012–2015 (10.2% against 4.3%), and is close to zero from 2016 (9.3% against 8.3%, then 4.9% against 4.7%). The convergence therefore comes in stages after 2008 rather than as a single break in that year.

Entity fixed effects move the levels but not the change. With them, the categorical Pre-SAA gap rises to +2.581 (p = .002), and the gaps after the SAA and after candidacy are +1.530 (p = .050) and +1.694 (p = .051). These levels are the same crossover artifact identified in Section 5.2 and carry no weight. The interaction, which measures the change, is −1.051 with entity fixed effects and −1.046 without them.

The candidacy test is the weakest in the design. The Post-candidacy period covers only 2022 and 2023, and post_candidacy equals 1 for all of 2022 although candidate status came in December. The coding biases the interaction toward zero and the short period leaves it little power (Section 4.8), so its null result says little either way. Overall, accession leaves the ethnic gap in FDI type unchanged. The one change appears in arrival. A modest Bosniak-majority advantage in attracting any FDI disappears after 2008. Section 5.4 tests whether this result survives the checks set out in Section 4.8.

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Table 6: *Type difference-in-differences, Bosnia*

	(1)	(2)	(3)	(4)
Bosniak share	+0.001	+0.003		
	(0.002)	(0.002)		
Bosniak-majority			+0.117	+0.334*
			(0.157)	(0.202)
Croat-majority			−0.187	+0.073
			(0.203)	(0.262)
Bosniak share × post-SAA	−0.002	−0.002		
	(0.002)	(0.002)		
Bosniak-majority × post-SAA			−0.204	−0.182
			(0.164)	(0.166)
Croat-majority × post-SAA			−0.003	−0.008
			(0.221)	(0.222)
Bosniak share × post-candidacy	−0.002	−0.002		
	(0.002)	(0.002)		
Bosniak-majority × post-candidacy			−0.143	−0.122
			(0.175)	(0.175)
Croat-majority × post-candidacy			+0.388**	+0.393**
			(0.173)	(0.173)
Log population	−0.127***	−0.133***	−0.121***	−0.121***
	(0.039)	(0.036)	(0.036)	(0.036)
Entity FE	No	Yes	No	Yes
Joint p, all interactions	.327	.325	.004	.009
Joint p, Bosniak interactions	—	—	.266	.372
N	201	201	201	201
R^2	0.080	0.109	0.103	0.121

OLS on active municipality-years, with export_share as the outcome. Standard errors in parentheses, clustered by municipality. Main terms give the Pre-SAA gap. The post-SAA and post-candidacy main effects are included but not shown. Bosniak share is in percentage points. Reference: Serb-majority. Joint p = Wald test that the listed interactions are zero. Croat-majority × post-candidacy rests on two municipality-years. *** p < .01, ** p < .05, * p < .10.

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Table 7: *Arrival difference-in-differences, Bosnia*

Panel A: Coefficients

	Logit				LPM			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bosniak share	+0.006	+0.006			+0.038	+0.056		
	(0.005)	(0.006)			(0.040)	(0.036)		
Bosniak-majority			+0.750*	+2.581***			+0.040	+0.062*
			(0.450)	(0.818)			(0.036)	(0.037)
Croat-majority			+0.324	+2.179**			−0.016	+0.007
			(0.606)	(0.992)			(0.035)	(0.038)
Bosniak share × post-SAA	−0.012***	−0.012***			−0.067***	−0.067***		
	(0.004)	(0.004)			(0.023)	(0.023)		
Bosniak-majority × post-SAA			−1.046***	−1.051***			−0.064***	−0.064***
			(0.354)	(0.358)			(0.020)	(0.020)
Croat-majority × post-SAA			−0.521	−0.530			−0.018	−0.018
			(0.537)	(0.542)			(0.021)	(0.021)
Bosniak share × post-candidacy	+0.005	+0.005			+0.037	+0.037		
	(0.008)	(0.008)			(0.044)	(0.044)		
Bosniak-majority × post-candi-dacy			+0.169	+0.164			+0.020	+0.020
			(0.551)	(0.556)			(0.035)	(0.035)
Croat-majority × post-candidacy			−0.365	−0.370			−0.015	−0.015
			(1.097)	(1.109)			(0.043)	(0.043)
Log population	+1.587***	+1.606***	+1.569***	+1.618***	+0.054***	+0.054***	+0.054***	+0.054***
	(0.209)	(0.222)	(0.215)	(0.222)	(0.014)	(0.015)	(0.014)	(0.014)
Entity FE	No	Yes	No	Yes	No	Yes	No	Yes
Joint p, all inter-actions	.024	.024	.052	.055	.009	.009	.024	.024
Joint p, Bosniak interactions	—	—	.013	.013	—	—	.004	.004
N	2,982	2,982	2,982	2,982	2,982	2,982	2,982	2,982
Pseudo R^2 / R^2	0.228	0.229	0.229	0.234	0.076	0.078	0.079	0.079

Panel B: Implied Bosniak–Serb gap by period (no entity FE)

Period	(1) Logit, share	(3) Logit, majority	(5) LPM, share	(7) LPM, majority
Pre-SAA	+0.006 (0.005)	+0.750* (0.450)	+0.038 (0.040)	+0.040 (0.036)
Post-SAA	−0.005 (0.005)	−0.296 (0.386)	−0.029 (0.033)	−0.023 (0.029)
Post-candidacy	0.000 (0.008)	−0.126 (0.610)	+0.008 (0.054)	−0.003 (0.046)

Outcome: has_fdi, on all municipality-years. Logit coefficients are log-odds. LPM coefficients are changes in the yearly probability of receiving FDI. In the logit, Bosniak share is in percentage points. In the LPM it is scaled ×100, so the coefficient is the effect of moving from 0 to 100% Bosniak. The post-SAA and post-candidacy main effects are included but not shown. Reference: Serb-majority. Panel B: gap = main effect plus the relevant interactions, with SE from the clustered covariance matrix. The Croat-majority terms fail the pre-trend test (Table B2) and are not read as DiD estimates. Standard errors in parentheses, clustered by municipality. *** p < .01, ** p < .05, * p < .10.

5.4 Robustness of the Arrival Result

The Bosniak × post-SAA interaction in the arrival model is the only significant interaction in the design that is not an artifact. Section 4.8 sets out the checks it must pass. Table 8 reports them for the four specifications without entity fixed effects, next to the full-sample estimate.

First, no single municipality drives the result. The jackknife re-estimates each model 142 times, dropping one municipality each time. In the continuous logit, the interaction stays between −0.013 and −0.011, and no p-value rises above .015. In the categorical logit, it stays between −1.256 and −0.942 (largest p = .008). The two LPM specifications are just as stable (Table 8).

Second, the five largest FDI destinations do not drive it either. Novo Sarajevo, Banja Luka, Grad Mostar, Tuzla, and Zenica hold about 58% of Bosnian projects and lie on both sides of the entity line. Dropping them leaves 137 municipalities. The interaction stays significant in all four specifications. In the logit it grows rather than shrinks, to −0.016 (p = .003) and −1.391 (p = .001), and in the LPM it does not move (−0.067 and −0.064).

Third, the placebo is null. Estimated on the Post-SAA years alone (2008–2021), the placebo milestone in 2015 returns no significant shift in any specification (lowest p = .143). The point estimates, however, are negative throughout, and their size ranges from about a third to three quarters of the main estimate. This fits the pattern in Table B1, where the convergence comes in stages rather than as a single break in 2008. The placebo also carries the caveat from Section 4.8 that 2015 is the year the SAA entered into force, so it is not an event-free date. The arrival result is therefore a difference between the years before and after 2008. It does not show that the agreement itself produced the change.

Fourth, and binding, the result cannot be separated from a shift between the two entities. Entity fixed effects remove level differences between the Federation and Republika Srpska, but not a change in that difference after a milestone. Table 8 therefore adds interactions between Republika Srpska and each milestone. The model must then separate ethnicity from entity using the six municipalities that cross the entity line. In the continuous logit, the ethnic interaction keeps its size (−0.014) but its standard error doubles, and it is no longer significant (p = .107). In the continuous LPM, it stays significant (−0.068, p = .035). In both, the Republika Srpska × post-SAA term is close to zero (p = .776 and p = .975). The categorical logit becomes unstable. It returns two large terms that offset each other, −4.317 for the ethnic interaction and −3.379 for the entity interaction, and is not informative. The data therefore cannot tell whether the convergence after 2008 runs along ethnic lines or along the entity line.

Taken together, the arrival result passes the influence and specification checks. It is not driven by a single municipality, by the largest destinations, or by functional form. Two limits remain. Its timing is less sharp than a single break in 2008, and it cannot be separated from a shift between the entities.

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Table 8: *Robustness of the Bosniak × post-SAA arrival interaction, Bosnia*

Check	Logit, share	Logit, majority	LPM, share	LPM, majority	N
Full sample	−0.012***	−1.046***	−0.067***	−0.064***	2,982
	(0.004)	(0.354)	(0.023)	(0.020)	
	p = .006	p = .003	p = .003	p = .001	
Jackknife, drop 1 of 142	−0.012	−1.046	−0.068	−0.064	2,961
	[−0.013, −0.011]	[−1.256, −0.942]	[−0.073, −0.058]	[−0.067, −0.055]	
	max p = .015	max p = .008	max p = .006	max p = .003	
Drop 5 FDI hubs	−0.016***	−1.391***	−0.067***	−0.064***	2,877
	(0.005)	(0.430)	(0.023)	(0.021)	
	p = .003	p = .001	p = .004	p = .002	
Placebo milestone 2015, 2008–2021 only	−0.009	−0.527	−0.032	−0.023	1,988
	(0.006)	(0.464)	(0.026)	(0.023)	
	p = .143	p = .257	p = .218	p = .311	
+ RS × milestone: ethnic × post-SAA	−0.014	−4.317**	−0.068**	−0.134*	2,982
	(0.009)	(1.809)	(0.032)	(0.078)	
	p = .107	p = .017	p = .035	p = .088	
+ RS × milestone: RS × post-SAA	−0.210	−3.379	−0.001	−0.078	2,982
	p = .776	p = .064	p = .975	p = .315	

Each cell reports the Bosniak × post-SAA interaction (Bosniak share or Bosniak-majority vs. Serb-majority) from Equation 3 without entity fixed effects. The exception is the last two blocks, which add Republika Srpska × post-SAA and × post-candidacy; for the RS term, only the coefficient and p-value are shown. Jackknife: median estimate, [minimum, maximum] across 142 refits, each on 2,961 municipality-years, and the largest p-value. Hubs: Novo Sarajevo, Banja Luka, Grad Mostar, Tuzla, Zenica. Placebo: false milestone at 2015, estimated on 2008–2021; 2015 is the year the SAA entered into force. LPM Bosniak share scaled ×100. Standard errors in parentheses, clustered by municipality. *** p < .01, ** p < .05, * p < .10.

5.5 Croatia as a Benchmark

Croatia serves as a benchmark, not a test (Section 3.5). Its Serb-majority municipalities, 18 under the 2011 census and 16 under the 2021 census, record no FDI in any year. There is therefore no ethnic gap to measure and no change in that gap across accession. As Section 3.5 noted, these municipalities lie largely in areas most affected by the war, so their lack of FDI cannot be read as an ethnic effect. What Croatia can show is how FDI in its Croat-majority municipalities developed through a completed accession. Table 9 reports this in two panels.

Panel A shows the raw picture. In Croat-majority municipalities, participation is the same before and after accession in 2013, at 2.6% of municipality-years. The number of projects rises from 322 to 432. However, the period after accession is one year longer (2013–2023 against 2003–2012), and active municipality-years rise only from 139 to 154. The export-oriented share edges down, from 0.168 to 0.137 of projects. Croatian FDI is also more market-seeking than Bosnian FDI. Across the whole period, 641 of its 754 projects, or 85%, are market-seeking, against 66% in Bosnia. This runs against the expectation set out in Section 1. The country that completed accession hosts the more market-seeking mix. Because the two panels are not pooled, this comparison is descriptive only. Croatian FDI is also concentrated. Zagreb alone holds 37% of Croatian projects (Section 4.4).

Panel B adds the controls. Once log population is held constant, accession changes neither margin. The post-accession coefficient is +0.116 for arrival (p = .428) and −0.030 for the export-

oriented share (p = .461). Completed accession thus leaves both arrival and type in Croat-majority municipalities unchanged.

For symmetry with Bosnia, Panel B also reports the continuous Serb share and its interaction with post-accession. Both interactions are null (0.000, p = .998; −0.002, p = .762). Without the interaction, the Serb share is null for arrival (−0.001, p = .889) and marginal for type (+0.011, p = .092). These terms are identified mainly off Serb minorities inside Croat-majority municipalities. They are not a comparison with Serb-majority municipalities and carry no weight for either hypothesis.

Croatia therefore offers a point of comparison, not a test. In a country that completed accession, FDI in Croat-majority municipalities neither arrived more often nor became more export-oriented. Because Croatia has no ethnic contrast, it cannot show whether accession would have changed one.

Table 9: *Croatia, the completed-accession benchmark*

Panel A: Participation and export-oriented share by period

Group	Period	Municipalities	Municipality-years	Active	Participation	Projects	Export-oriented	Pooled share
Croat-majority	Pre-accession	538	5,380	139	0.026	322	54	0.168
	Post-accession	540	5,932	154	0.026	432	59	0.137
Serb-majority	Pre-accession	18	180	0	0.000	0	0	—
	Post-accession	18	184	0	0.000	0	0	—

Panel B: Estimates

Model	Part	Term	Coefficient (SE)	p	N
Croat-majority municipalities	Arrival (logit)	Post-accession	+0.116 (0.147)	.428	11,312
	Type (OLS)	Post-accession	−0.030 (0.041)	.461	293
All municipalities	Arrival (logit)	Serb share × post-accession	0.000 (0.014)	.998	11,676
	Type (OLS)	Serb share × post-accession	−0.002 (0.008)	.762	293
All municipalities, no interaction	Arrival (logit)	Serb share	−0.001 (0.010)	.889	11,676
	Type (OLS)	Serb share	+0.011* (0.006)	.092	293

Pre-accession 2003–2012; post-accession 2013–2023. Majority group from the 2011 census for 2003–2016 and the 2021 census for 2017–2023 (Croatian Bureau of Statistics 2012, 2022), so municipality counts differ slightly between periods. Panel A counts a municipality once per period in which it holds that majority; 16 municipalities are Serb-majority under the 2021 census. Panel B: control is log population; no entity effects; type models on active municipality-years only. Serb share in percentage points. The Serb-share terms rest mainly on Serb minorities inside Croat-majority municipalities. Standard errors in parentheses, clustered by municipality. *** p < .01, ** p < .05, * p < .10.

6 Discussion

This section interprets the results in four steps. It covers the type of FDI, the role of city size, the change in arrival after 2008, and the overlap between ethnicity and entity, which limits how far that change can be read.

6.1 Ethnic Composition and FDI Type

The first hypothesis expected the ethnic signal to show in the type of FDI. It does not. Where FDI arrives, Bosniak-majority municipalities attract no higher export-oriented share than Serb-majority ones, and the raw shares run the other way (Section 5.1). The null has a clear reach. Without entity fixed effects, it rules out a Bosniak-majority advantage of more than about 0.16 in the export-oriented share. It does not rule out a modest one, because the type model rests on only 201 municipality-years. It also describes FDI only where it arrives (Section 4.6).

Two readings fit this result. First, export-oriented investors may read the signal at the national level rather than the municipal one. Their sensitivity to the environment (Pandya 2016, 457, 469) would then decide whether they choose Bosnia, not where in Bosnia they locate. Second, there is little for the signal to sort. Only 121 of Bosnia's 356 projects are export-oriented, so a preference among these investors would have to be large to move a municipal share. Both readings are consistent with the null, and the design tests neither.

Nor does the null settle the alternative channel from Section 3.3. If investors favor municipalities whose political orientation matches that of their home country, ethnic composition would shape FDI type through who invests rather than through how investors read risk (Ledyaeva, Karhunen, and Kosonen 2013). The two channels could also pull in opposite directions and offset each other. Separating them requires the origin of each project's capital, which the fDi Markets records contain but this paper does not use.

6.2 City Size, Arrival, and Type

City size explains the one ethnic difference the raw data shows in arrival. Bosniak-majority municipalities receive FDI more often than Serb-majority ones, but the difference disappears once log population is held constant (Section 5.2). They are larger, and larger municipalities draw investment more often. The participation gap is therefore a matter of size, not of ethnic composition, and it has no bearing on the first hypothesis, which concerns type.

Size pulls the two margins in opposite directions. Larger municipalities receive FDI more often, yet the FDI they receive is less often export-oriented. The three largest destinations, one per group, show this most clearly. Novo Sarajevo received 5 export-oriented projects of 102, Banja Luka 13 of 52, and Grad Mostar 1 of 20. This pattern is consistent with the logic of the two types. Market-seeking investment follows fixed features such as the size of the market (Pandya 2016, 457), so it gathers where consumers are. Export-oriented production is freer to locate elsewhere (Pandya 2016, 457, 469) and has less reason to favor the largest places.

The design cannot separate size from what comes with it. Novo Sarajevo lies within the capital, and Banja Luka is the administrative center of Republika Srpska. Log population captures these functions as well. The pattern may therefore describe the largest destinations above all, and this paper does not test whether it holds among smaller municipalities.

6.3 Whether FDI Arrives, Not What Kind

The one change across accession lies in whether FDI arrives, not in what kind arrives. After 2008, the ethnic gap in the export-oriented share did not change, while a modest Bosniak-majority advantage in arrival disappeared without turning into a disadvantage (Section 5.3).

For the two directions set out in Section 3.4, the type results matter most, because the second hypothesis concerns type. Nothing points toward a strengthening of the signal. The Bosniak–Serb interaction with post-SAA is negative, and without entity fixed effects its 95% confidence interval rules out a widening of the gap by more than about 0.12 (Table 6). A weakening, by contrast, cannot be judged, because before 2008 there was no measurable type gap to narrow (Table B3). The design could detect the first direction but not the second.

The arrival result runs in the direction the weakening reading predicts, in which national commitments that bind the whole state and are monitored from outside may stand in for the local signal (Pandya 2016, 469). Two other readings fit equally well. The convergence may reflect a shift between the two entities (Section 6.4), or a gradual drift unrelated to any milestone, since it comes in stages and the 2015 placebo points the same way, although it is not significant (Section 5.4). The change is therefore dated after 2008, not attributed to the SAA, and none of the three readings supports the second hypothesis.

Three limits from Section 4.8 also bear on these readings. The pre-trend tests rest on few observations, so they support parallel trends only weakly. The candidacy test covers two years and counts all of 2022 as post-candidacy, so it says little either way. Finally, the Bosnian ethnic shares come from a single, disputed census, which adds error to every ethnic estimate.

Croatia offers a point of comparison. Across a completed accession, its Croat-majority municipalities received FDI neither more often nor in a more export-oriented mix (Section 5.5). Because Croatia has no ethnic contrast, it cannot show whether accession moves an ethnic gap.

6.4 Ethnicity or Entity

The arrival result cannot be read along ethnic lines alone. Sixty of the 64 Serb-majority municipalities lie in Republika Srpska, and only six municipalities cross the entity line (Section 4.8). Once the models allow the two entities to drift apart after 2008, the ethnic interaction keeps its size but loses significance in the continuous logit, though not in the LPM. The entity interaction stays close to zero, and the categorical logit becomes unstable (Section 5.4). The data cannot tell the two readings apart, and each deserves equal weight.

On the ethnic reading, investors read the composition of a municipality, and that reading changed after 2008. On the entity reading, the change ran through the Federation and Republika Srpska themselves. They are hard to separate even in principle, because the political contest in Bosnia is “purposefully contained within ethnic and subnational boundaries” (Kapidžić 2020, 81). Most of the evidence for the signal in Section 3.2 also describes the entities, not individual municipalities. The signal the argument attributes to a municipality may therefore be one that investors read at the level of the entity.

The same overlap widens the confidence interval of the type estimates with entity fixed effects (Section 5.2). Separating ethnicity from entity requires variation that six crossover municipalities cannot supply.

7 Conclusion

This paper has asked whether the ethnic composition of municipalities in Croatia and Bosnia predicts the composition of inward FDI, and whether that relationship changes as the two countries move through EU accession. The evidence supports neither hypothesis. Where FDI

arrives in Bosnia, Bosniak-majority municipalities attract no higher export-oriented share than Serb-majority ones, nor did this change as accession advanced. In Croatia, a completed accession left Croat-majority municipalities unchanged in both whether FDI arrives and what kind. The one change lies in arrival. Before 2008, Bosniak-majority municipalities held a modest advantage in attracting any FDI at equal size; after 2008 it disappears, and the data cannot tell whether this change runs along ethnic or entity lines. The type results rule out large effects, not modest ones.

This paper addresses a question that the literatures reviewed in Section 2 leave open. For Bosnia, the answer is that ethnic composition does not sort FDI by type at the municipal level. The two margins also behave differently. Larger municipalities receive FDI more often but a less export-oriented mix, and the only change after 2008 lies in arrival. A study of volume alone would have seen only the arrival margin, where a Bosniak-majority advantage fades after 2008, and would have missed that no such difference appears in the type of FDI. The results are also consistent with investors reading political signals above the municipality, at the level of the state or the entity (Sections 6.1 and 6.4). In practice, they give investment-promotion bodies no reason to expect a different mix of FDI from the ethnic composition of a municipality.

Three extensions follow from what this design left open. First, the fDi Markets records include each investor's home country. Using that information would separate how investors read risk from who invests, the channel that Section 3.3 named but did not test. Second, tests of accession as a moderator need a gap that exists before the milestone. Where there is none, as for FDI type in Bosnia, a narrowing cannot be observed, and only one of the two directions in Section 3.4 can be tested. Third, whether the link between city size and FDI type holds beyond the largest destinations remains open.

Finally, the overlap between ethnicity and entity in Bosnia (Section 6.4) calls for a setting in which the two come apart. Serbia offers one. It signed its SAA in the same year as Bosnia (Stabilisation and Association Agreement with Serbia 2008), and, unlike Bosnia, it contains municipalities with different ethnic majorities within a single jurisdiction, across Sandžak and within the autonomous province of Vojvodina (Statistical Office of the Republic of Serbia 2023, table 2). Comparing them holds the jurisdiction fixed while ethnic composition varies. Which group reads as EU-aligned is less obvious there than in Bosnia, so the case is valuable for its design rather than for a predicted result, and my master's thesis takes it up. More broadly, the question for subnational research on FDI is not only whether investors read political signals, but at which level they read them, whether the state, the jurisdiction, or the municipality.

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Appendix A: FDI Classification Key

This appendix lists the sector key used in Section 4.2. Each of the 37 fDi Markets sectors that appear in the data is assigned to one type. In a second step, projects in export-oriented sectors whose business activity is Retail or Sales, Marketing & Support are recoded as market-seeking. The last column reports how many projects that step moved.

Table A1: *Sector key and project counts*

Sector	Bosnia	Croatia	Recoded to market-seeking
Export-oriented sectors (24)			
Aerospace	0	5	1
Automotive Components	17	6	0
Automotive OEM	5	3	1
Beverages	2	5	0
Biotechnology	1	1	0
Business Machines & Equipment	0	6	3
Ceramics & Glass	1	2	1
Chemicals	8	8	6
Consumer Electronics	4	7	9
Electronic Components	7	9	1
Engines & Turbines	0	1	0
Food, Beverages & Tobacco	29	62	56
Industrial Machinery, Equipment & Tools	15	17	6
Medical Devices	1	2	1
Metals	12	11	1
Minerals	3	1	0
Non-Automotive Transport OEM	0	2	1
Paper, Printing & Packaging	2	2	0
Pharmaceuticals	13	20	20
Plastics	6	9	0
Rubber	1	2	0
Semiconductors	0	1	0
Textiles	62	193	232
Wood Products	5	4	0
Subtotal	**194**	**379**	**339**
Market-seeking sectors (13)			
Alternative/Renewable Energy	13	20	—
Building & Construction Materials	12	8	—
Business Services	15	27	—
Coal, Oil & Natural Gas	7	2	—
Communications	10	19	—
Consumer Products	20	80	—
Financial Services	30	29	—
Healthcare	3	1	—
Hotels & Tourism	3	31	—
Leisure & Entertainment	2	2	—
Real Estate	17	65	—
Software & IT Services	17	45	—
Transportation, Warehousing & Storage	13	46	—
Subtotal	**162**	**375**	**—**
Total	**356**	**754**	**339**

Located projects, 2003–2023. Counts by sector are before the activity rule. Of the 339 recoded projects, 73 are in Bosnia and 266 in Croatia. After recoding, 121 Bosnian and 113 Croatian projects are export-oriented (Table 1). The activity rule applies only to export-oriented sectors, so the last column is empty for market-seeking sectors.

Appendix B: Additional Results

This appendix reports the additional results referred to in Sections 5.3 and 5.4. Table B1 shows participation and the export-oriented share by group before each milestone. Table B2 reports the pre-trend tests. Table B3 reports the implied ethnic gap in each period for every specification.

Table B1: *Participation and export-oriented share before each milestone, Bosnia*

Window	Bosniak participation	Bosniak share (projects)	Croat participation	Croat share (projects)	Serb participation	Serb share (projects)
2003	0.167	0.294 (17)	0.042	0.000 (2)	0.031	0.333 (6)
2004	0.111	0.091 (11)	0.000	— (0)	0.063	0.400 (5)
2005	0.130	0.313 (16)	0.042	0.000 (1)	0.031	0.250 (4)
2006	0.093	0.375 (8)	0.083	0.000 (2)	0.031	0.000 (3)
2007	0.148	0.647 (17)	0.083	0.500 (2)	0.031	0.333 (3)
2008–2011	0.079	0.286 (42)	0.010	0.000 (1)	0.047	0.474 (19)
2012–2015	0.102	0.250 (40)	0.031	0.091 (11)	0.043	0.625 (16)
2016–2018	0.093	0.417 (24)	0.097	0.571 (7)	0.083	0.409 (22)
2019–2021	0.049	0.056 (18)	0.069	0.200 (5)	0.047	0.250 (20)

Participation = share of municipality-years with at least one project. Share = pooled export-oriented share, with the number of projects in parentheses. Descriptive only; most cells rest on very few projects.

Table B2: *Pre-trend tests (ethnic group × linear year trend, pre-milestone years only)*

Window	Outcome	Term	Coefficient (SE)	p	N
Pre-SAA, 2003–2007	Arrival	Bosniak share × trend	−0.001 (0.003)	.824	710
	Arrival	Bosniak-majority × trend	+0.053 (0.209)	.799	710
	Arrival	Croat-majority × trend	+0.546** (0.276)	.048	710
	Type	Bosniak share × trend	+0.001 (0.001)	.361	53
	Type	Bosniak-majority × trend	+0.097 (0.106)	.362	53
	Type	Croat-majority × trend	+0.078 (0.138)	.568	53
Post-SAA, 2008–2021	Arrival	Bosniak share × trend	−0.001 (0.001)	.220	1,988
	Arrival	Bosniak-majority × trend	−0.053 (0.056)	.345	1,988
	Arrival	Croat-majority × trend	+0.155** (0.063)	.014	1,988
	Type	Bosniak share × trend	0.000 (0.000)	.791	126
	Type	Bosniak-majority × trend	+0.002 (0.030)	.941	126
	Type	Croat-majority × trend	+0.015 (0.038)	.691	126

Arrival = logit; type = OLS on active municipality-years. Control: log population. Reference: Serb-majority. SE clustered by municipality. *** p < .01, ** p < .05, * p < .10.

Table B3: *Implied ethnic gap by period, all specifications*

Model	Contrast	Entity FE	Pre-SAA	Post-SAA	Post-candidacy
Arrival, logit	Bosniak share	No	+0.006 (0.005)	−0.005 (0.005)	0.000 (0.008)
		Yes	+0.006 (0.006)	−0.006 (0.005)	0.000 (0.008)
	Bosniak vs Serb	No	+0.750* (0.450)	−0.296 (0.386)	−0.126 (0.610)
		Yes	+2.581*** (0.818)	+1.530** (0.779)	+1.694* (0.869)
	Croat vs Serb	No	+0.324 (0.606)	−0.197 (0.375)	−0.562 (0.971)
		Yes	+2.179** (0.992)	+1.649* (0.894)	+1.279 (1.222)
Arrival, LPM	Bosniak share	No	+0.038 (0.040)	−0.029 (0.033)	+0.008 (0.054)
		Yes	+0.056 (0.036)	−0.011 (0.027)	+0.025 (0.049)
	Bosniak vs Serb	No	+0.040 (0.036)	−0.023 (0.029)	−0.003 (0.046)
		Yes	+0.062* (0.037)	−0.002 (0.027)	+0.018 (0.043)
	Croat vs Serb	No	−0.016 (0.035)	−0.034 (0.026)	−0.049 (0.042)
		Yes	+0.007 (0.038)	−0.011 (0.030)	−0.026 (0.043)
Type, OLS	Bosniak share	No	+0.001 (0.002)	−0.001 (0.001)	−0.003 (0.002)
		Yes	+0.003 (0.002)	+0.001 (0.002)	−0.001 (0.003)
	Bosniak vs Serb	No	+0.117 (0.157)	−0.087 (0.124)	−0.231 (0.197)
		Yes	+0.334* (0.202)	+0.151 (0.180)	+0.030 (0.244)
	Croat vs Serb	No	−0.187 (0.203)	−0.190 (0.135)	+0.198 (0.139)
		Yes	+0.073 (0.262)	+0.065 (0.213)	+0.459** (0.215)

Gap = main effect plus the relevant interactions, with SE from the clustered covariance matrix. LPM Bosniak share scaled ×100. The entity-FE categorical arrival gaps are identified off the six crossover municipalities (Section 5.2). The Croat type gap after candidacy rests on two municipality-years. *** p < .01, ** p < .05, * p < .10.

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